7

Transport of Calcium, Magnesium, and Phosphate

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CHAPTER OUTLINE

KEY POINTS

- The renal tubular transport of minerals is under the tight regulation of key hormonal systems, including parathyroid hormone, 1,25-dihydroxyvitamin D, and fibroblast growth factor 23 (FGF-23). These and other hormonal systems influence the reabsorption of calcium, magnesium, and phosphate to varying degrees in a segment-specific manner. Furthermore, a number of drugs and the calcium ion itself impact tubular transport processes, leading to significant alterations in mineral balance.
- While paracellular transport of calcium in the proximal tubule is predominantly facilitated by claudin-2 and claudin-12, respectively, the paracellular reabsorption of calcium and magnesium in the thick ascending limb occurs through the claudin-16/19 complex. This latter permeation can be inhibited by claudin-14, which is tightly regulated by the tubular calcium-sensing receptor. Pathogenic variants in several of these claudins can lead to calcium wasting and kidney stone disease.
- Phosphate transport primarily occurs in the proximal tubule via a transcellular pathway mediated by the apical transporters SLC34A1 (Napi-2a) and SLC34A3 (Napi-2c). Transport depends on the regulated trafficking and expression of these transporters in the brush border membrane, and pathogenic variants have marked effects on mineral balance.
- Transcellular transport of calcium and magnesium takes place in the distal nephron via epithelial channels TRPV5 and TRPM6, respectively. The transport of both calcium and magnesium in this segment is regulated by a number of proteins and hormonal systems.

THE ROLE OF CALCIUM IN THE BODY

Calcium is an essential mineral and the most abundant cation in the human body.[1-3] In a 70-kg man, the total body content typically averages around 1 kg, with the majority (99%) stored in bone.[2] Calcium is critical to many physiologic processes throughout the body including the contraction of muscle, intracellular signaling, blood clotting, gene transcription, and exocytosis of hormones and neurotransmitters, and it is integral to bone stability.[4] As these important physiologic processes depend on the presence of this mineral, calcium imbalances can result in a number of pathologic consequences. Therefore there is considerable regulation to maintain the calcium concentration in blood. Approximately 45% of the calcium in blood is found in its free ionized form, which is also the form with biological activity. A smaller portion, 10% to 15%, is complexed with anions like phosphate, sulfate, and citrate, while the remaining, approximately 40% to 45%, is bound to proteins, primarily albumin[5-7] (Fig. 7.1). Changes in blood pH affect ionized calcium levels by altering the albumin-bound fraction. Free ionized calcium in blood is maintained within a narrow physiologic range, between 1.10 and 1.35 mM (4.4–5.4 mg/dL), and the total concentration ranges between 2.2 to 2.6 mM (8.8 to 10.3 mg/dL) in healthy adult subjects.[7-9] The blood calcium level tends to slightly higher in children and, as such, age-appropriate normal ranges should be used in children.[10] The inability to effectively regulate blood calcium concentrations within the normal range can lead to a number of sequelae. Hypocalcemia can result in a variety of symptoms including tetany, carpopedal spasms, and life-threatening arrhythmias.[11] In contrast, hypercalcemia can cause symptoms including fatigue, polyuria, nephrolithiasis, nausea, mood alterations, coma, and abnormal heart rhythms, which can ultimately lead to cardiac arrest[12] (see Chapter 17 for extended description of calcium disorders). Calcium balance is controlled via a concerted interplay between the kidneys, the intestine, and the bone, whereby intestinal absorption, mobilization out of bone and the reabsorption along the nephron can be adjusted to stabilize calcium levels in the desired range (Fig. 7.2).

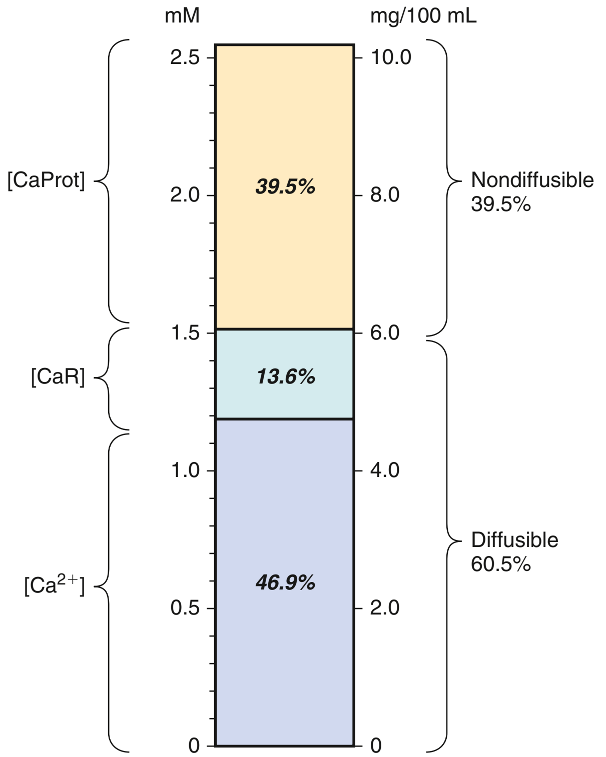

Fig. 7.1 Components of serum total calcium assessed by ultrafiltration data in normal human patients. *Ca^{2+},* Ionized calcium; *CaProt,* protein-bound calcium; *CaR,* diffusible calcium complexes. (Redrawn from Moore EW. Ionized calcium in normal serum, ultrafiltrates, and whole blood determined by ion-exchange electrodes. *J Clin Invest*. 1970;49:318–334, with permission of the publisher.)

TRANSPORT OF CALCIUM

Adult humans typically consume around 1000 mg of calcium, with approximately 400 mg of this being absorbed along the intestine[13] (see Fig. 7.2). In the growing child, as well as during pregnancy, this varies significantly, with a much greater fractional absorption occurring in these physiologic states.[14,15] The absorption of dietary calcium occurs through either a passive paracellular pathway or via active transcellular mechanisms.[16,17] The dominant route for calcium absorption is via the passive paracellular route; however, this requires a sufficiently high concentration of calcium in the intestinal lumen and hence adequate intake of dietary calcium.[18] Importantly, if ingestion of calcium is low, this can lead to a negative calcium balance, which over time will negatively impact bone mineralization.[19] Osteoblasts and osteoclasts, specialized cells responsible for bone formation and resorption, respectively, facilitate the exchange of calcium between extracellular fluid and bone.[20] The kidneys are critical to the maintenance of calcium balance and act by amending the transport activity across the tubular epithelia, whereby they determine how much of the filtered calcium is returned back into the circulation and how much is excreted in urine.

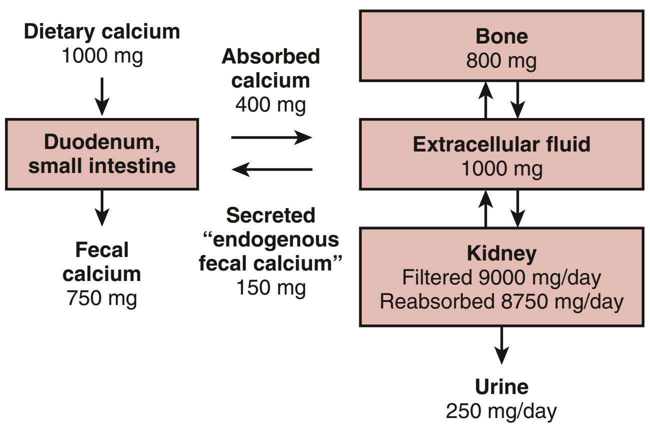

Fig. 7.2 Maintenance of calcium homeostasis in healthy adults involves the regulation of calcium absorption in the intestine, exchange with bone, and subsequent reabsorption by the kidney.

Table 7.1 Distribution and Concentrations of Magnesium in a Healthy Adult

Site	% Total-Body Mg	Concentration/Content
Bone	53	0.5% of bone ash
Muscle	27	9 mmol/kg wet weight
Soft tissue	19	9 mmol/kg wet weight
Adipose tissue	0.012	0.8 mmol/kg wet weight
Erythrocytes	0.5	1.65-2.73 mmol/L
Serum	0.3	0.69-0.94 mmol/L

From Yu ASL, Chertow GM, Luyckx V, Marsden PA, Skorecki K, Taal MW, eds. *Brenner and Rector's The Kidney*. 11th ed. Philadelphia, PA: Elsevier; 2019.

THE ROLE OF MAGNESIUM IN THE BODY

Magnesium is the fourth most abundant cation in the body, which is predominantly stored in soft tissue and bone[1-3,21-23] (Table 7.1). Magnesium functions as a necessary cofactor in a large number of enzymatic processes, such as those involved in glycolysis and oxidative phosphorylation, either by direct protein binding or in complex with ATP substrates.[24-26] Furthermore, magnesium is a direct modulator of ion channel activity.[27] As such, failure to maintain magnesium balance can lead to a range of clinical symptoms to be discussed later. Magnesium can be found in plasma in three different forms: Approximately 55% exists in a free ionized form, roughly 30% is bound to plasma proteins, and the remainder is complexed with anions such as phosphate and citrate.[6] Blood magnesium levels range from 0.75 to 0.96 mM (1.82–2.33 mg/dL) in adults.[23] Children tend to have higher blood magnesium levels, and as such age appropriate levels should be employed.[28] Hypomagnesemia can cause a plethora of symptoms including neuromuscular irritability, leading to tetany and seizures, as well as cardiac arrhythmias. Patients with hypomagnesemia often experience secondary hypocalcemia, which adds complexity to their clinical assessment and treatment.[29] Hypermagnesemia can lead to a range of symptoms such as lethargy, coma, and in extreme situations, cardiac arrest.[29] See Chapter 17 for additional information on hypomagnesemic and hypermagnesemic conditions).

Magnesium levels in plasma represent <1% of total body magnesium. Surrogate markers of cellular magnesium content have been identified, such as measuring muscle or erythrocyte magnesium content, as well as assessing the ability to excrete magnesium loads by the kidney.[21,22,30] These methods can be cumbersome and are not routinely performed outside experimental settings. Furthermore, insufficient details are available regarding the equilibrium between magnesium pools, and the fraction of ionized magnesium in either tissue or blood is rarely determined. In patients with altered magnesium balance, there was a direct correlation between serum magnesium levels and both bone and erythrocyte magnesium content, whereas the magnesium content in muscles did not consistently correlate with serum magnesium concentration.[31] Overall, blood magnesium levels are assumed to reflect whole body magnesium amounts to some degree.

TRANSPORT OF MAGNESIUM

As with calcium, the intestinal absorption of magnesium is the major route of uptake into the body. The average daily intake of dietary magnesium is about 300 mg in adults.[32] Nevertheless, absorption of magnesium can vary greatly and ranges from 11% to 65% depending on the amount of dietary magnesium ingested.[33] Magnesium can be absorbed through passive paracellular or active transcellular transport mechanisms along the intestine.[33-35] The paracellular route predominates when dietary intake of magnesium is adequate, while the active transcellular route facilitates absorption when the concentration of magnesium in the intestinal lumen is low.[36] Magnesium is also an important component of bone, accounting for approximately half of the total magnesium in the body.[30] In rats, bone magnesium content is reduced when their diet is low in magnesium.[37] Furthermore, hypomagnesemia in rats leads to increased bone remodeling, reduced bone mineral density, and bone strength.[38-41] Consequently, magnesium may be deposited and resorbed from bone in a comparable way to calcium, yet the existence of a regulatory mechanism mediating the mobilization of magnesium from bone remains unidentified. Further, the degree to which magnesium deficiency contributes to osteoporosis in humans remains to be clarified.[42] Again, the kidneys play a critical role in regulating magnesium balance, amending the urinary magnesium excretion to maintain blood magnesium levels, processes we know much more about than the deposition and absorption of magnesium from bone.

THE ROLE OF PHOSPHATE IN THE BODY

Phosphate is the most abundant anion in the body and plays a crucial role in multiple biological processes. Phosphate constitutes an integral part of bone as a component of the hydroxyapatite crystal. Furthermore, phosphate is critical for numerous other physiologic processes, where it is required for cellular energy reserves as a constituent of ATP and nucleotides, as part of the backbone of DNA and RNA, as a central component involved in the control of cellular signaling by protein phosphorylation, and as a buffer.[43,44] Approximately 85% of phosphate is stored as part of hydroxyapatite in bone, whereas the remaining phosphate is predominantly stored inside the cells. Most of the intracellular phosphate is either bound and contained in phospholipids in cell membranes, present as inorganic phosphate esters, or phosphorylated molecules and proteins.[43,44] Only a small fraction, <1%, of total body phosphate is present in a soluble form in the extracellular fluid. In blood, 10% of inorganic phosphate is bound to proteins, 5% is complexed with other ions such as calcium, magnesium, and sodium, whereas the remaining 85% exists in ionic forms as either monohydrogen phosphate (HPO_4^{2-}) or dihydrogen phosphate ($H_2PO_4^-$).[45] The proportion of each is determined by the pH and at pH 7.4 exists as a ratio of approximately 4:1.[44] Under physiologic conditions, serum phosphate is maintained in the range of 0.80 to 1.45 mmol/l (2.5–4.5 mg/dL) in adults.[45] Children have higher plasma phosphate levels, necessitating the use of an age-appropriate normal range.[46] Hypophosphatemia can result acutely in fatigue and muscle weakness. More chronic depletion of plasma phosphate contributes to poorly mineralized bones and the syndrome of rickets in children and osteomalacia in adults. In contrast, acute hyperphosphatemia is typically asymptomatic, though chronic increases in plasma phosphate, as occurs in chronic kidney disease (CKD) and end-stage kidney failure (ESKF) can result in the formation of calcium phosphate precipitation throughout the body, leading to end-organ dysfunction and cardiovascular disease. Additional information on conditions with hyperphosphatemia and hypophosphatemia can be found in Chapter 17.

TRANSPORT OF PHOSPHATE

Daily intake of phosphorus amounts to some 800 to 1400 mg[45] in an adult. As with the divalent cations, intestinal absorption of phosphate involves at least two separate uptake mechanisms, namely paracellular phosphate transport across the bowel, which relies on the electrochemical gradients and active transport facilitated by sodium-dependent phosphate cotransporters.[47-50] Being an integral part of hydroxyapatite, which is the major mineral of bone, phosphate is deposited and resorbed via the concerted actions of osteoblasts and osteoclasts, the same as for calcium. The kidneys play a crucial role in regulating systemic phosphate levels, as they are responsible for eliminating phosphate in urine or increasing the reclamation of phosphate back into blood when needed.

OVERALL HOMEOSTATIC REGULATION OF MINERAL BALANCE

The main regulatory mechanisms maintaining mineral balance are primarily directed at controlling calcium and phosphate, whereas homeostatic mechanisms to regulate blood magnesium have not been well defined. Blood calcium is maintained within a narrow physiologic range. This is facilitated by the seven transmembrane G-protein coupled calcium-sensing receptor (CaSR), which is highly expressed on the surface of parathyroid chief cells[51] (Fig. 7.3A). Here it detects increments in plasma calcium and responds by inhibiting parathyroid hormone (PTH) expression and secretion.[51,52] The CaSR-mediated inhibition of PTH secretion is blunted when plasma calcium levels are lowered, allowing PTH to be secreted into the bloodstream.[51,53] Furthermore, phosphate appears to act directly on the CaSR to stimulate secretion of PTH.[54] PTH drives resorption of bone via its receptor, PTH1R, situated on the osteoblast. The activation of PTH1R by PTH on the osteoblast stimulates osteoclasts to break down hydroxyapatite and osteoid, thereby releasing calcium and phosphate into the circulation[55,56] (see Fig. 7.3A). PTH1R is also expressed in renal tubular epithelial cells where binding to PTH increases urinary phosphate excretion and decreases urinary calcium excretion.[57-59] PTH also stimulates the production of the active form of the hormone 1,25-dihydroxyvitamin D (calcitriol). It does so by increasing the expression of CYP27B1 (1-α hydroxylase) in the proximal tubule, which catalyzes the 1-α hydroxylation of 25-hydroxyvitamin D into 1,25-dihydroxyvitamin D[60] (see Fig. 7.38). 1,25-dihydroxyvitamin D acts to stimulate intestinal calcium and phosphate absorption, the release of calcium and phosphate from bone, and the reabsorption of calcium and phosphate from the kidney.[61] The combined actions of 1,25-dihydroxyvitamin D and PTH raise plasma calcium while maintaining or reducing plasma phosphate levels depending on dietary intake. The other phosphocalciotropic hormone, FGF23, is produced and secreted by osteocytes and osteoblasts in response to either increased blood phosphate levels or active 1,25-dihydroxyvitamin D.[62] Activation of FGF receptors relies on its cofactor, klotho. FGF23 increases urinary phosphate excretion, reduces 1,25-dihydroxyvitamin D formation, and enhances the renal reabsorption of calcium[62,63] (see Fig. 7.3C). Therefore similar to PTH and active 1,25-dihydroxyvitamin D, FGF23 controls calcium and phosphate balance.

TRANSPORT OF CALCIUM, MAGNESIUM AND PHOSPHATE IN THE KIDNEY

CALCIUM

Calcium exists in protein-bound and unbound fractions in blood with only the unbound fraction able to be filtered at the glomerulus. This results in 9 to 10 g of calcium filtered by the kidneys each day (see Fig. 7.2). Only about 1% of the filtered calcium is excreted in the urine because the bulk of it is reabsorbed throughout the nephron.[64-66] The filtered calcium is reabsorbed through paracellular transport pathways in the proximal tubule (PT) and the thick ascending limb (TAL) and via the transcellular pathway along the distal convolution (encompassing the distal convoluted tubule and connecting tubule). As such, the PT contributes the majority (~65%) of calcium reabsorption from the filtrate,[67,68] while the TAL accounts for approximately 25% of the calcium reabsorption[69] and the distal convolution the remaining 4% to 8%.[70-72] The collecting duct does not significantly reabsorb calcium (Fig. 7.4).

MAGNESIUM

Seventy percent of the magnesium in blood is freely filtered (i.e., the unbound fraction)[73] and, as such, the magnesium concentration in the initial filtrate after passing the glomeruli is 0.5 mM.[64,71] Of the magnesium filtered by the kidney, 3% to 5% is excreted into urine[74] (see Fig. 7.4). In comparison with calcium, only approximately 25% of the filtered magnesium is reabsorbed via the paracellular route in the PT.[75-78] Consequently, the bulk of magnesium reabsorption takes place via the paracellular pathway in the TAL, where 60% to 65% of the filtered magnesium is reabsorbed.[79,80] The final site for magnesium reabsorption in the kidney is the distal convoluted tubule, where 5% to 6% of the filtered magnesium is reabsorbed through a transcellular pathway[71,75,80,81] (see Fig. 7.4).

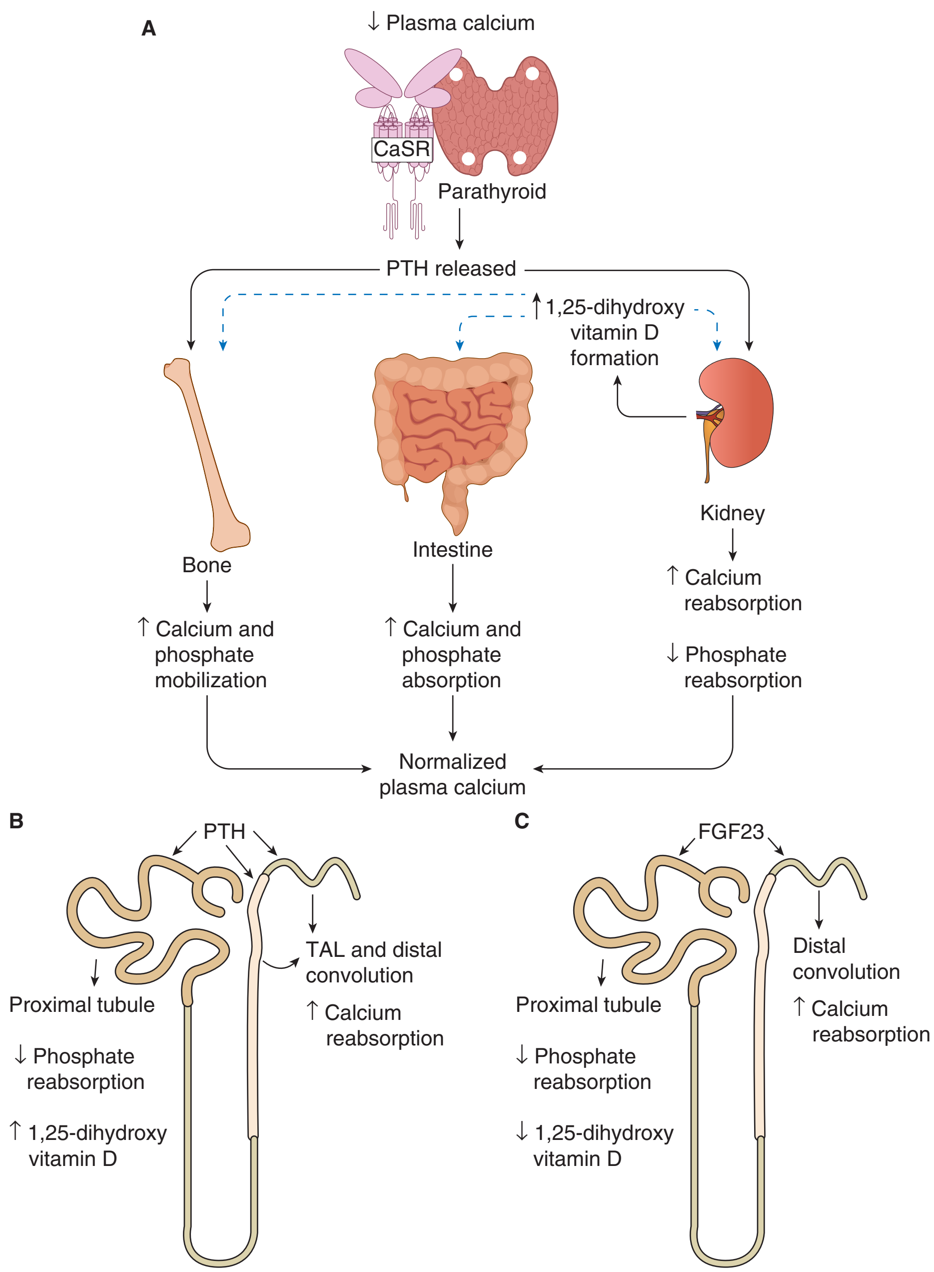

Fig. 7.3 The parathyroid hormone–vitamin D axis in maintenance of calcium balance. Parathyroid hormone (PTH), 1,25-dihydroxyvitamin D3 (active vitamin D), and fibroblast growth factor 23 (FGF23) are involved in the regulation of renal tubular calcium and phosphate transport. (A) Release of PTH from the parathyroid glands is stimulated by low levels of plasma calcium. PTH induces bone resorption, leading to the release of calcium and phosphate into the plasma. PTH enhances urinary calcium reabsorption and phosphate excretion in the kidney. In the parathyroid, PTH stimulates 1-α-hydroxylase to convert 25-hydroxyvitamin D3 into 1,25-dihydroxyvitamin D. The presence of active vitamin D enhances the absorption of calcium and phosphate from the intestine. (B) PTH has multiple effects on the kidney. In the proximal tubule, it reduces phosphate reabsorption and stimulates the production of 1,25-dihydroxyvitamin D. PTH also inhibits NHE3-mediated sodium transport in this segment, which may reduce proximal tubular calcium reabsorption. In the thick ascending limb (TAL) and distal convoluted tubule, PTH enhances calcium reabsorption. Overall, the net effect of PTH is increased renal calcium retention. (C) High 1,25-dihydroxyvitamin D levels and phosphate promote the secretion of FGF23 from bone. FGF23 functions as a negative feedback regulator of 1,25-dihydroxyvitamin D, increases the reabsorption of calcium from the distal nephron, and decreases reabsorption of phosphate from the proximal tubule.

PHOSPHATE

As only 10% of the phosphate in blood is protein bound, the remaining 90% is freely filtered at the glomerulus. Phosphate is reabsorbed through tubular transport mechanisms, allowing 80% to 90% to be returned back into the circulation with the difference, 10% to 20%, excreted in the urine. Nearly all phosphate reclamation occurs along the proximal convoluted and straight tubules.[82-86] Some authors suggest an additional distal reclamation site, but the existence of this pathway is debated[83] (see Fig. 7.4).

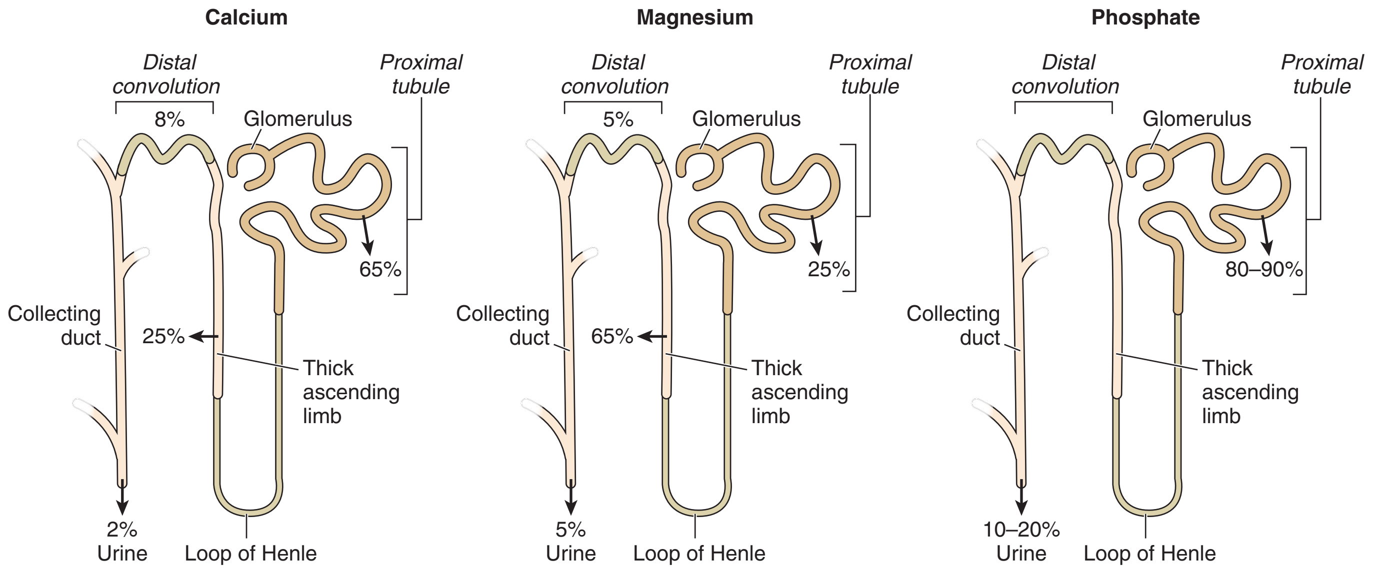

Fig. 7.4 Percentages of filtered calcium, magnesium, and phosphate reabsorbed along the renal tubule.

TRANSPORT OF CALCIUM, MAGNESIUM, AND PHOSPHATE IN THE PROXIMAL TUBULE

CALCIUM

Approximately 65% of the calcium filtered by the kidney is reabsorbed along the segments of the PT, largely via passive paracellular transport.[67,87-89] Up to 10% of calcium absorption in the straight portion of the proximal tubule may occur by the transcellular pathway, via a voltage-gated calcium channel-dependent mechanism.[90] PT calcium loss contributes to the development of hypercalciuria and kidney stone formation,[91,92] underscoring the importance of understanding the molecular pathways responsible for calcium reclamation in the PT.

The reabsorption of sodium and resulting reabsorption of water driven by the osmotic force generated are the primary mechanisms allowing the transport of calcium across the PT.[67] Sodium entry into the PT is primarily driven via the sodium-hydrogen exchanger isoform 3 (NHE3), encoded by the *SLC9A3* gene. This sodium transport is facilitated by an inward-directed sodium gradient, which is generated by sodium efflux from the cell by the basolaterally expressed Na^+-K^+-ATPase[93] (Fig. 7.5). The sodium and subsequent water reclamation facilitates passive paracellular calcium reabsorption by creating a calcium concentration gradient from the tubular lumen to the blood. Aquaporin-1 water channels are highly expressed in the apical and basolateral membranes of the PT.[94,95] Consequently, transcellular water reabsorption predominates in the PT (~70%), but there is also considerable reabsorption of water via the paracellular pathway.[96,97] This paracellular flux may promote the process of solvent drag, which is the convective movement of calcium through the tight junction.[68,98] Calcium transport through the tight junction of the PT may be facilitated by either the calcium concentration gradient or solvent drag, and the relative contribution of each has not been established. Consistent with this, micropuncture data show a close relationship between sodium, fluid, and calcium reabsorption.[87,99]

The paracellular reabsorption of divalent cations may also be impacted by the transepithelial voltage gradient across the PT. In the early PT, a small lumen negative potential difference of about –2 mV is generated by sodium-coupled glucose reabsorption.[100] This transepithelial voltage gradient gradually turns positive along the length of the PT (~ +2 mV), due to the preferential paracellular reabsorption of chloride.[101] The relative ion selectivity of the proximal tubular tight junction pores also changes along the length of this segment. In the early PT, the pore is slightly cation selective, whereas in the late PT, the pore is anion selective.[102] These changes may contribute to the shift in charge of the transepithelial voltage gradient along the PT.[101] In the absence of experimental data, modeling studies have been completed on calcium transport across the PT, which suggest that the transepithelial voltage gradient has a marginal contribution to the transport of calcium across the segment.[98]

The PT is a leaky epithelium with a transepithelial resistance of only 5 to 15 $\Omega \cdot cm^2$.[103,104] The permeability of the PT, as in other epithelia, is conferred by the claudin family. Claudins are four-pass transmembrane proteins. Their expression in the tight junction allows them to interact in both the same cell (cis-interaction) and between neighboring cells (trans-interaction). This facilitates the formation of pores or barriers between cells that determines the permeability of the paracellular shunt, thereby enabling the paracellular movement of specific ions.[105] In adult animals, the PT expresses claudin (CLDN)-2, CLDN12, and CLDN10a,[103,106,107] with CLDN2 and CLDN12 contributing to the formation of a cation permeable pore, whereas CLDN10a forms an anion permeable pore.[101,103,108-110]

PT permeability to calcium is contributed by CLDN2 and CLDN12[110] (see Fig. 7.5). When overexpressed in cell culture, permeation of calcium is increased by CLDN2 in monolayers from both MDCK I, OK cells, and CaCo2 cells determined using either radioactive flux studies or bi-ionic diffusion potential measurements.[110-112] Both sodium, calcium, and water permeate the paracellular shunt via CLDN2,[112,113] and water flux can be driven by sodium through this pathway.[114,115] It is not yet clear whether similar electrolyte and water flux can occur through CLDN12. Nevertheless, CLDN12 increases paracellular calcium permeability when expressed in CaCo2 cells, as determined

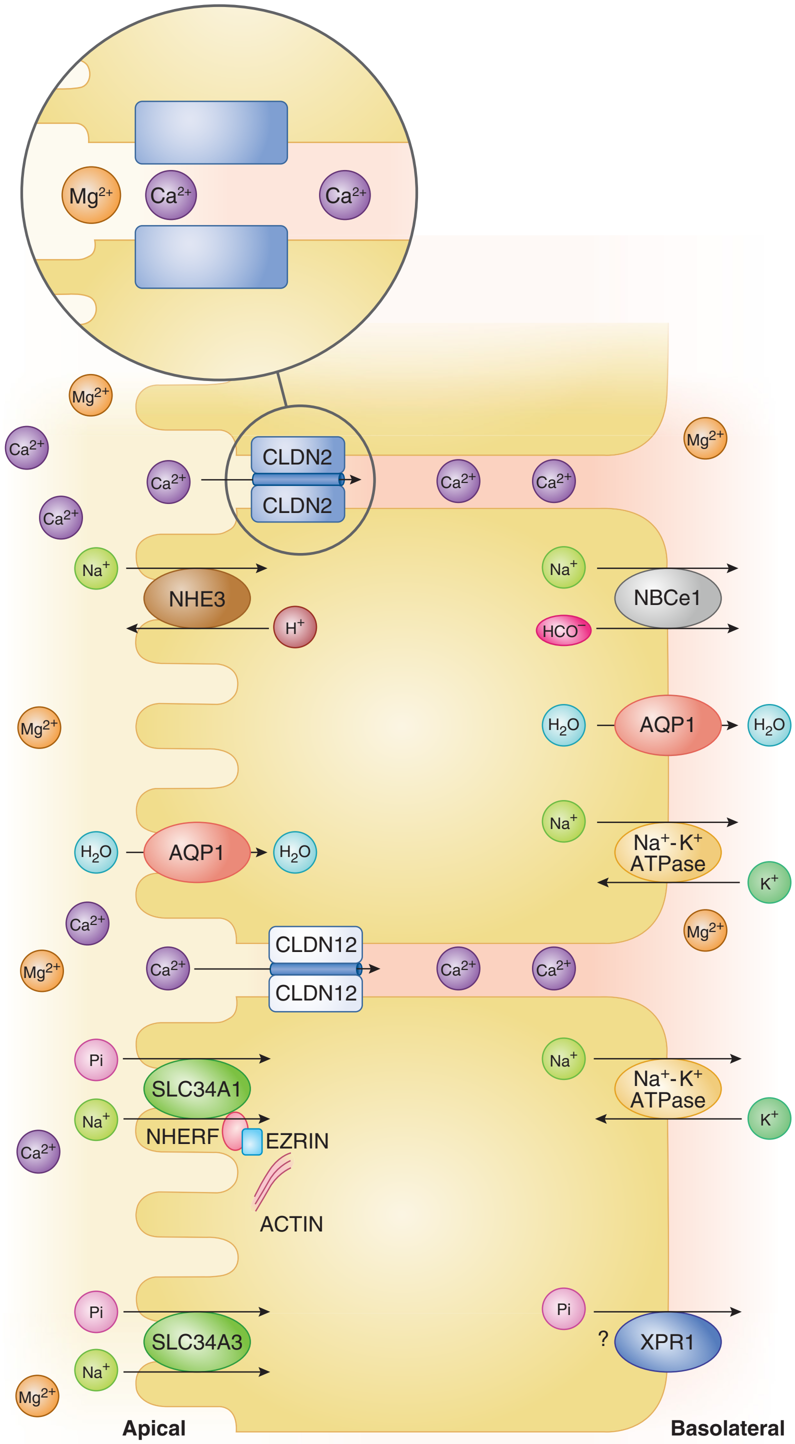

Fig. 7.5 Mechanisms by which calcium, magnesium, and phosphate are transported in the proximal tubule (PT). Within the proximal tubule, the sodium proton exchanger isoform 3 (NHE3) facilitates the inward transport of sodium, while the sodium-potassium-ATPase (Na^+-K^+-ATPase) is responsible for the efflux of sodium. Additionally, the electrogenic sodium bicarbonate cotransporter (NBCe1) is responsible for the efflux of sodium and bicarbonate. Paracellular calcium reabsorption is facilitated by the reabsorption of sodium and water across the PT and occurs primarily through claudin-2 (CLDN2) and also claudin-12 (CLDN12). Paracellular magnesium permeation is less than calcium in the PT. The insert serves as a magnified representation of the tight junction, indicating the preferential permeability of calcium to magnesium in the PT. For the sake of clarity, paracellular sodium and water fluxes have been omitted. The process of phosphate uptake is facilitated by the SLC34A1 and SLC34A3 transporters, driven by the sodium gradient. SLC34A1 is retained on the brush border membrane by NHERF-1. Ezrin promotes the binding of NHERF-1 to actin. The xenotropic and polytropic retroviral receptor (Xpr1) may contribute in the process of phosphate extrusion at the basolateral membrane.

by measurements of bi-ionic diffusion potential and radioactive calcium fluxes.[110,111]

Only the PT and initial part of the thin descending limb in long looped nephrons express CLDN2.[106] *Cldn2*-deficient mice exhibit both hypercalciuria and nephrocalcinosis.[108,109] Although *Cldn12*-deficient mice *do not* exhibit hypercalciuria, ex vivo microperfusion of PTs from this strain demonstrates that CLDN12 confers permeability to both calcium and sodium in the PT as well.[103] Data from *Cldn2* and *Cldn12* double knockout mice provide more evidence for the significance of CLDN12 in PT calcium transport.[110] Compared with *Cldn2*-deficient mice, these animals show exaggerated renal calcium wasting leading to hypocalcemia and decreased bone mineral density.[108,110] CLDN2 and CLDN12 form independent, partially redundant paracellular pores, each contributing to calcium permeability.[110]

The most common metabolic abnormality associated with kidney stone disease is hypercalciuria.[89,116] Decreased calcium reabsorption from PT is a mechanism causing hypercalciuria in some cases.[91,92,108] In line with these findings is the observation that *Slc9a3* null animals exhibit elevated urinary calcium excretion.[117] Children with pathogenic variants in *SLC9A3* experience sodium-losing diarrhea, but the excretion of calcium in urine has not been investigated.[118,119] Furthermore, single nucleotide polymorphisms (SNPs) in *AQP1* correlate with stone disease in humans, underscoring the importance of the reclamation of water to the reabsorption of calcium in the PT.[120] However, the precise mechanism by which these SNPs contribute to an increased risk of stone formation is currently unknown. Noncoding SNPs in the *CLDN2* gene that associate with either increased or reduced expression of CLDN2 correlate with reduced or increased risk of kidney stone disease, respectively. Furthermore, in an Iranian family, kidney stones appear to be caused by coding pathogenic variants in *CLDN2*.[108] SNPs in CLDN12 have not been linked to the development of kidney stones; however, this is not unexpected considering that *Cldn12* knockout mice do not exhibit hypercalciuria.[103]

MAGNESIUM

As with calcium, magnesium is reabsorbed from the proximal tubule via the paracellular route (see Fig. 7.5). Detailed micropuncture studies demonstrate that the flux of magnesium across the PT increases in a nonsaturating manner as luminal magnesium concentrations rise. Furthermore, when punctured tubules are perfused with a magnesium-free solution, backflux of magnesium occurs.[79,121] However, in comparison with calcium, only up to 30% of the filtered magnesium is reclaimed from the PT in adult animals[75,78-80] (see Fig. 7.4). The kinetics of paracellular magnesium transport differ markedly from those of both sodium and calcium. In fact, the magnesium concentration in the lumen needs to rise markedly (TF/UF >1.9) for uptake of magnesium to occur.[76,80] This is in line with findings from flux studies, which suggested that the PT permeability to magnesium is substantially lower than that of calcium (1×10^{-5} cm/s for magnesium, $7–27 \times 10^{-5}$ cm/s for calcium),[122-124] although a direct comparison of permeabilities was not performed. It is of interest to note that younger animals display higher PT magnesium reabsorption than adults.[125]

Since the two main cation-permeable claudins expressed in the adult PT are CLDN2 and CLDN12,[107] these may allow magnesium permeation across the epithelium. Consistent with this, magnesium permeability was higher in MDCK cell monolayers when *CLDN2* was overexpressed.[101] However, *Cldn2*-deficient mice or *Cldn2* and *Cldn12* double-deficient mice do not exhibit either hypomagnesemia or urinary magnesium wasting.[103,108-110] This may be explained by the comparatively small amount of magnesium transport in the PT and the ability of both the TAL and distal convolution to compensate PT losses. In fact, following deletion of *Cldn10a,* the junctional patches that would have expressed CLDN10a along the PT now become populated by CLDN2. These *Cldn10a*-deficient animals do display both lower urinary magnesium excretion and hypermagnesemia, as well as a higher relative permeability for magnesium in the S1 and S3 segments.[101] *Cldn6* and *Cldn9* expression has been documented in the PT of younger animals with higher PT magnesium transport,[107] though they are unlikely to confer permeability to magnesium since these claudins are barrier forming.[126] Ultimately, the identity of the PT magnesium pore needs to be clearly delineated.

PHOSPHATE

The PT is the main site of phosphate reabsorption, reclaiming at least 80% of the filtered load.[127] There is significant variation in the transport of phosphate along the length of the PT. Detailed micropuncture measurements have determined that the majority of filtered phosphate is reclaimed in the convoluted portion of the PT, with the bulk of this in the early convoluted portion.[128,129] In the PT, phosphate influx is largely transcellular and mediated by the sodium-coupled phosphate transporters expressed in this segment. These include the sodium-dependent phosphate cotransporter 2A (NaPi-2a, SLC34A1) encoded by the *SLC34A1* gene, the sodium-dependent phosphate cotransporter 2C (NaPi-2c, *SLC34A3*) encoded by the *SLC34A3* gene, and PIT2 encoded by the *SLC20A2* gene.[130-132] *SLC34A1* and *SLC34A3* mediate most, if not all, of the phosphate reclamation in the PT[133] (see Fig. 7.5), and a role for PIT2 remains to be established. Importantly, the *SLC34* transporters are most abundantly expressed in the proximal convoluted tubule, where the highest reclamation of phosphate is observed.[134,135] *SLC34A1* drives electrogenic sodium and phosphate cotransport[136,137] with a stoichiometry of 1 divalent phosphate ion to 3 sodium ions.[138] In contrast, *SLC34A3* is electroneutral and transports 1 divalent phosphate ion and 2 sodium ions.[139]

SLC34A1 is the main isoform in mice, accounting for the majority of phosphate reclamation by the PT.[133,140,141] Renal-specific deletion of *Slc34a3* in mice does not affect renal phosphate transport.[142] This may result from SLC34A1 compensating for this loss. Consistent with this hypothesis, double *Slc34a1*-and *Slc34a3*-deficient mice display significantly reduced plasma phosphate and elevated fractional phosphate excretion, in comparison with *Slc34a1*-deficient animals. In humans, *SLC34A3* may have a more prominent role, as inactivating pathogenic variants in *SLC34A3* cause the rare disease hereditary hypophosphatemic rickets with hypercalciuria (HHRH), a syndrome with renal phosphate wasting and high 1,25-dihydroxyvitamin D levels, with the latter augmenting intestinal absorption of calcium and causing hypercalciuria.[143,144] Inactivating pathogenic variants in *SLC34A1* can cause idiopathic infantile hypercalcemia, a rare childhood disease. Affected patients display renal phosphate wasting and hypophosphatemia at presentation.[145,146] In *Slc34a1*-deficient mice, and potentially also patients, the hypercalcemia results from suppression of FGF23 due to lowered phosphate levels and subsequent enhanced synthesis of 1,25-dihydroxyvitamin D.[146] SLC20A2 does not appear important for PT phosphate transport, despite being present in the apical membrane and its expression being altered in response to changes in dietary phosphate intake.[147] As such, patients with pathogenic variants in the *SLC20A2* gene exhibit idiopathic basal ganglia calcification but do not have alterations in phosphate homeostasis.[148,149]

A number of interactions between the SLC34A1 transporter and binding proteins have been described to occur via its C-terminal PDZ domain[150,151] including with the sodium/hydrogen exchanger regulatory factor (NHERF) family,[152] which are expressed in or near the brush border membrane[150,153] (see Fig. 7.5). The inability to properly interact with NHERF1 via the PDZ domain impairs the proper expression of SLC34A1 in the brush border membrane, leading to disassembly, subsequent endocytosis of SLC34A1, and eventual degradation in lysosomes.[154] This is exemplified in mice with target deletion of the *Slc9a3r1* gene encoding NHERF1. These animals display a loss of SLC34A1 from the brush border, hypophosphatemia, and renal phosphate wasting.[155] Furthermore, genetic variants found in the human *NHERF1* gene are associated with nephrolithiasis or bone demineralization and were identified only in patients with low tubular phosphate reclamation.[156] While NHERF3 shows a similar interaction, phosphate handling in the proximal tubule and SLC34A1 regulation seem marginally affected in *Slc9a3r3*-deficient mice.[157] In contrast, NHERF3 may play a role in maintaining SLC34A3 in the brush border membrane. When wild-type and *Slc9a3r3*-deficient mice were maintained on a low-phosphate diet, wild-type animals displayed an increased SLC34A1 and SLC34A3 abundance at the brush border, while only SLC34A1 appeared to increase to the wild-type level in *Slc9a3r3* deficient mice.[158] Phosphate exits across the basolateral membrane. Although a potential contribution from the xenotropic and polytropic retroviral receptor 1 (XPR1) has been described, the basolateral efflux mechanisms for phosphate in the PT are still not fully delineated[159] (see Fig. 7.5).

TRANSPORT OF CALCIUM, MAGNESIUM, AND PHOSPHATE IN THE THICK ASCENDING LIMB

CALCIUM AND MAGNESIUM

Available studies suggest that calcium and magnesium are reclaimed from the filtrate via the same paracellular pathway in the TAL, and therefore the transport of both electrolytes is discussed together in this section. Micropuncture studies have demonstrated that the TAL is responsible for reabsorbing up to 60% of magnesium filtered by the glomerulus and approximately 25% of filtered calcium[69,72,160] (see Fig. 7.4). This is not surprising given that reclamation of calcium is higher than for magnesium in the PT. Reclamation of calcium and magnesium in the TAL is largely paracellular. Some reports indicate the presence of a transcellular transport pathway for calcium in the TAL; however, the contribution of this to TAL calcium transport and the identity of the pathway are not well understood.[161-163]

Paracellular calcium and magnesium reabsorption is passive and therefore dependent on the electrochemical gradients. This is illustrated in perfusion experiments of isolated segments from cortical TAL, where applying a lumen-negative voltage facilitates diffusion of divalent cations into the lumen of the tubule, in comparison with directional transport from lumen to bath under normal conditions.[164] The majority of paracellular divalent cation transport takes place in the cortical TAL and outer stripe of outer medulla (OSOM), whereas limited calcium and magnesium reclamation occurs from the inner stripe of outer medulla (ISOM).[162,165-169] This is largely due to the permeability characteristics of the shunt, as is discussed in detail later.

The lumen-positive potential ranges from +5 to +13 mV in the early part of the medullary TAL and rises as the segment ascends such that in the cortical TAL the lumen-positive potential may reach up to +30 mV.[170-174]

Divalent cation reclamation is highly dependent on the lumen-positive electrochemical gradient across the TAL.[164,175] The transepithelial voltage gradient is formed by the asymmetric transport of sodium, potassium, and chloride across the TAL epithelium. NKCC2, the furosemide-sensitive cotransporter, drives transport of sodium, potassium, and two chloride ions from the lumen into the epithelial cell,[171,176,177] with sodium being effluxed by the Na^+-K^+-ATPase at the basolateral surface. ROMK channels in the apical membrane allow the recycling of potassium back into the tubular lumen,[171,178-180] whereas CLC-K channels and the Barttin subunit, which is necessary for CLC-K channel function, allow the exit of chloride ions across the basolateral membrane[181-184] (see Chapter 6) (Fig. 7.6). Loop diuretics act by potently inhibiting NKCC2 transport,[185,186] which effectively collapses the voltage gradient and hence the transepithelial movement of divalent cations via the paracellular shunt.[164,175,187,188]

Loop diuretics, such as furosemide, are primarily used in pediatric and adult patient populations for the treatment of edema. As they reduce the transepithelial voltage gradient in the TAL, they also limit the reabsorption of divalent cations. The results of this are observed in experimental animal models and humans treated with loop diuretics, which cause urinary calcium and magnesium wasting.[187,189-191]

Pathogenic variants in the *SLC12A1* gene encoding NKCC2, the *KCNJ1* gene encoding ROMK, the *CLCNK2* gene encoding CLC-Kb (CLC-K2 in mice), or *BSDN* encoding Barttin cause various types of Bartter syndrome that strongly inhibit or abolish TAL electrolyte transport and hence the lumen-positive voltage critical for paracellular divalent cation reclamation.[176,178,181,192-194] Hypotension, hypokalemic metabolic alkalosis, excessive renal urinary sodium and chloride excretion, and hyperreninemia are hallmark features of Bartter syndrome.[195-197] The wasting of calcium and magnesium into the urine is also observed in Bartter syndrome, but the severity and affected divalent cation differ between genetic defects.

Pathogenic variants in *SLC12A1* and *KCNJ1* cause the antenatal forms of Bartter syndrome that present with polyhydramnios, which is frequently accompanied by premature birth. Furthermore, affected individuals

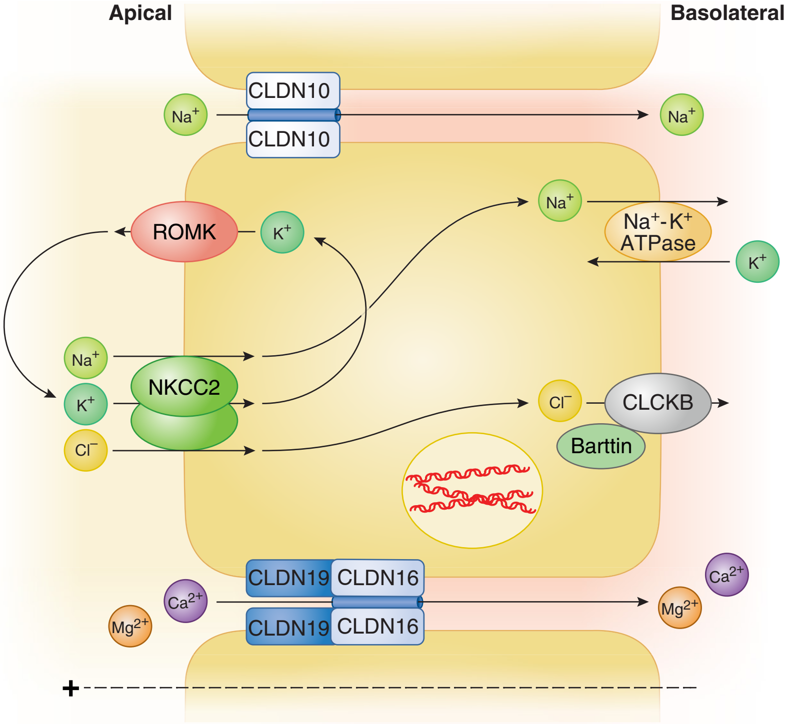

Fig. 7.6 Mechanisms by which calcium and magnesium are transported in the thick ascending limb (TAL). In normocalcemic conditions, the TAL exhibits asymmetric electrolyte secretion, resulting in the generation of a lumen-positive voltage that facilitates the reabsorption of calcium and magnesium via the paracellular shunt. This requires NKCC2 (furosemide-sensitive sodium, potassium, 2 chloride cotransporter), the renal outer medulla potassium channel (ROMK), and the chloride channel Kb (ClC-Kb) with its subunit Barttin, and the sodium-potassium-ATPase (Na^+-K^+-ATPase). The interaction between CLDN16 and CLDN19 creates a cation-permeable pore that allows for both paracellular calcium and magnesium reabsorption. CLDN10 forms a monovalent cation pore driving sodium reabsorption; however, the transtubular potential difference can be further augmented by the backflux of sodium toward the top of the TAL.

experience severe hypercalciuria and early-onset nephrocalcinosis, while hypomagnesemia and hypermagnesuria are not commonly observed in antenatal Bartter syndrome.[176,178,197,198] A newly identified transient form of antenatal Bartter syndrome caused by pathogenic variants in *MAGED2* also presents with hypercalciuria.[199] MAGED2 appears important for regulating NKCC2 expression, which leads to this antenatal Bartter form. However, the syndrome, including hypercalciuria, resolves spontaneously as the infant ages, though nephrocalcinosis may persist.[199] Pathogenic variants in the *CLCNKB* gene cause the classical form of Bartter syndrome. Here, larger phenotypic variability is seen at presentation, with a number of less severe cases. As such, hypercalciuria and nephrocalcinosis are less frequently observed, whereas hypomagnesemia develops frequently.[200] The presence of other efflux mechanisms enabling chloride exit from the TAL epithelial cell may explain why calcium wasting is not frequently encountered in patients with classical Bartter syndrome. Furthermore, the expression of CLC-Kb in the distal convoluted tubule (DCT) could influence the development of hypomagnesemia.[183,193] *Clcnk2*-deficient mice do not display hypercalciuria consistent with the traditional Bartter phenotype, which may result from the retained expression of the CLC-K1 chloride channel (CLC-Ka in humans), although this is predominantly found in the medullary TAL.[201] Increased urinary magnesium excretion was also found in these mice, which may be caused by reduced transcellular magnesium reabsorption from the DCT rather than the TAL since *Clcnk2* deletion causes atrophy of the early DCT segment.[201]

The TAL is a rather leaky epithelium with a transepithelial resistance ranging from 4 to 34 $\Omega \cdot cm^2$,[169,202-205] which is comparatively higher in the cortical segments.[169,172,206] Hence the transepithelial resistance found in the TAL is comparable to the PT.[104,109,207] The TAL exhibits significant variation of permeability characteristics along its length. As such, the permeability of sodium (PNa) may exceed that of chloride (PCl) by 2-fold to 10-fold.[168,169,173,206] In the TAL there is a higher PNa/PCl ratio in the ISOM that decreases toward the cortical TAL, while the opposite is found for calcium (PCa) and magnesium (PMg) permeabilities, with both divalent cation permeabilities being comparably higher as the TAL ascends toward the cortex.[168,169] Limited data are available, and direct comparisons of PCa and PMg are lacking from flux experiments under normal conditions, in part due to the lack of radioactive tracers for magnesium and suitable selective dyes that are only sensitive to magnesium (discussed in detail here[208]). Bi-ionic diffusion potential measurements in isolated mouse TALs, which require the establishment of large concentrations across the epithelial layer, suggest that PCa exceeds that of PMg.[168,169,209] The rate of paracellular transport does not depend on the permeability characteristics of the epithelium alone but on the electrochemical gradients. However, as discussed elsewhere,[208] direct measurements of the concentration of calcium and magnesium delivered to the loop bend have not been systematically performed, and the scarcity of data poses a challenge in determining the apparent differences. Furthermore, concentration measurements of cations at various segments in the TAL and peritubular capillaries at the same site would be needed but are not easily obtained. Overall, divalent cation permeability is highest along the OSOM and the cortical TAL.

The expression of select claudins and their restriction to specific tight junctions in the TAL likely accounts for the axial heterogeneity of the relative permeability differences observed for monovalent and divalent cations in the TAL. As such, while all cells in the TAL appear to express NKCC2, a number of distinct cell types have been identified by single-cell RNA sequencing,[210-212] which may enable them to contribute unique transport characteristics.[205] In human and mouse kidneys, two distinct cell populations can be differentiated by their expression of *CLDN16* and *CLDN10*.[211,212] CLDN16 and CLDN19 are primarily responsible for permeability to calcium and magnesium and hence the movement of divalent cations across the TAL. In contrast to the anion-selective CLDN10a predominantly expressed in the PT, it is the cation-selective CLDN10b that predominates in the TAL. CLDN10b is therefore important for the paracellular movement of sodium and other monovalent cations across the TAL epithelium.[213] Although this distribution of isoform expression is well established from expression studies in various *Cldn10*-deficient mouse strains,[101,213] currently available antibodies recognize epitopes in both, making them unsuitable for discerning their relative expression. It is assumed that the CLDN10 recognized in the TAL is primarily detecting the CLDN10b variant.

Both CLDN16 and CLDN19 are expressed in a subset of TAL cells, where they colocalize to the basolateral membrane domains and tight junction[168,214-218] (see Fig. 7.6). CLDN16 expression is concentrated to TAL segments in the OSOM and cortex, whereas CLDN19 is also found in the ISOM. However, there, it is not expressed in the tight junction.[168,217,219] Tight junctions not occupied by CLDN16 and CLDN19 in the OSOM and cortex are populated by CLDN10, giving rise to a mosaic expression pattern in the junctions along the length of these segments[168,217-220] (see Fig. 7.6). In addition, CLDN10 is abundantly expressed in the basolateral membranes and tight junctions of the ISOM. It is the junctional localization of the claudins that most likely determines the permeability characteristics of the epithelium. Overexpression of claudins in cell monolayers has produced variable results with respect to permeability characteristics, especially for CLDN16 and CLDN19. The reasons for that remains unclear but could relate to whether the protein reaches the tight junction or its interaction with claudins already expressed in the cell model (discussed in detail by Gunzel and Yu[221]). In mice, a detailed study linked the segmental permeability characteristics of the TAL epithelium with the junctional localization of claudins in the same tubule. Here, perfused cortical TAL tubules showed a three times higher PMg/PNa permeability ratio compared with tubules from the ISOM, which coincided with a higher abundance of CLDN16 in the tight junction.[168] Furthermore, the junctions mainly composed of CLDN10, which are highly abundant in the ISOM, where CLDN16 is absent, showed a higher PNa/PCl ratio, in line with CLDN10 forming a sodium-permeable pore, whereas CLDN16 and CLDN19 contribute to the formation of a divalent cation-permeable pore.[168]

These observations are well supported by human disease, where pathogenic variants in the *CLDN16* and *CLDN19* genes cause the autosomal recessive disorder of familial hypomagnesemia with hypercalciuria and nephrocalcinosis (FHHNC).[214,215] Loss of function of either gene causes significant urinary losses of divalent cations and, as a consequence, the development of hypomagnesemia and nephrocalcinosis, which can lead to renal insufficiency.[214,215,222,223] In contrast, pathogenic variations in *CLDN10* cause the HELIX syndrome, which is a rare autosomal recessive disorder characterized by Hypohidrosis, Electrolyte imbalance, Lacrimal gland dysfunction, Ichthyosis, and Xerostomia.[224,225] These patients present with a sodium chloride–losing phenotype with higher fractional excretion of sodium chloride. Furthermore, these patients display hypermagnesemia, which appears more pronounced in children compared with adults and is accompanied by inappropriately normal urinary magnesium excretion. Furthermore, urinary calcium excretion was normal, but blunted fractional calcium excretion was observed after furosemide injection.[224]

The reason for these findings can be explained on the basis of a detailed analysis of the tubular transport in animals lacking the *Cldn16* and *Cldn10* genes, or both. Mice with targeted deletion of *Cldn16* recapitulated parts of the FHHNC phenotype, displaying excessive losses of urinary calcium and magnesium but without nephrocalcinosis.[220,226] Furthermore, microperfused cortical TAL tubules from *Cldn16*-deficient mice displayed decreased permeability ratios of PCa/PNa and PMg/PNa, whereas no effect of gene deletion was observed on the permeability ratio of PNa/PCl, in line with CLDN16 forming a calcium and magnesium permeable paracellular pore. Divalent cation permeability ratios remained unchanged in perfused TALs from the medullary portion, where CLDN16 is largely absent.[220,226] Deficient *Cldn19* mice have been generated, but the permeability characteristics of the TAL have not been determined. However, in these animals, CLDN19 is required to keep CLDN16 in the tight junction and loss of *Cldn19* thus causes the removal of both from the junction, while CLDN16 is retained in the basolateral membrane.[219,227]

Mice with targeted deletion of *Cldn10* in the distal nephron including the TAL recapitulate some features of HELIX syndrome, with higher magnesium concentrations in blood. These mice also display decreased fractional magnesium excretion and marginally decreased fractional calcium excretion, in line with TAL hyperabsorption of these ions. Furthermore, *Cldn10*-deletion facilitated the development of nephrocalcinosis, which is absent in patients.[213,224,225] In perfused tubule experiments, distal tubular deletion of *Cldn10* leads to a reduction in the PNa/PCl in the TAL but with a concomitant increase in the permeability ratios of PCa/PNa and PMg/PNa. Furthermore, the transepithelial voltage gradient is markedly increased in *Cldn10*-deficient animals.[213] The high PCa/PNa and PMg/PNa in *Cldn10*-deficient animals result from repopulation of the CLDN10-deficient junctions with CLDN16, while the opposite is not seen in *Cldn16*-deficient animals.[213,220] In combination with the higher transepithelial voltage gradient following *Cldn10* deletion, this is a key driver for hyperabsorption of divalent cations along the TAL. In fact, *Cldn10*- and *Cldn16*-double-deficient mice display reduced PCa/PNa and PMg/PNa in perfused TAL from the cortex in comparison with Cldn10-deficient mice.[220] The development of nephrocalcinosis was also absent in double-deficient animals. Whether the same claudin redistribution occurs in the human kidney remains to be determined in detail for CLDN16, but CLDN19 was found redistributed to junctions in the TAL of a patient with mutations in *CLDN10*.[228] In addition to these pore-forming claudins that predominate in the TAL is the pore-blocking CLDN14, which is largely absent during baseline conditions but strongly stimulated by the CaSR (see section on CaSR effects on the kidney later).

PHOSPHATE

While phosphate is not reclaimed along the TAL, it is notable that the sodium-phosphate cotransporter SLC34A2 has been found to be expressed in the TAL. Its contribution to phosphate transport or other cellular mechanisms remains to be defined.[229]

TRANSPORT OF CALCIUM, MAGNESIUM, AND PHOSPHATE IN THE DISTAL CONVOLUTION

The distal convolution is critical for the transcellular reclamation of both calcium and magnesium. The distal convolution consists of the DCT, connecting tubule (CNT), and superficial portion of the cortical collecting duct (CD).[230] This subdivision is based on both morphology and the select expression profile of transport proteins.[212,230-232] Briefly, and as outlined in Chapter 6, NCC is a thiazide-sensitive NaCl cotransporter, encoded by the *SLC12A3* gene, expressed by all DCT cells. However, the segment can be further subdivided into an early (DCT1) and a late (DCT2) segment with the latter expressing the epithelial sodium channel (ENaC)[231,233] (Fig. 7.7). Additionally, throughout the rest of the collecting system, including the CNT and CD segments, which contribute to the distal convolution, both ENaC and Aquaporin 2 are expressed.[231,233,234] Furthermore, ROMK is expressed in the distal convolution and is important for potassium secretion, especially in the CNT and CD.[235-237] The transepithelial voltage across this segment is between 0 and –5 mV in the initial DCT and decreases in the CNT to –40 mV, in line with the shift from electroneutral sodium chloride reabsorption to electrogenic transport.[238-242] Because of the relatively higher transport capacity for sodium and potassium in the CNT, the transepithelial voltage gradient is substantially larger and decreases again to –15 mV toward the cortical CD.[243-246] The distal convolution generates a high epithelial resistance, ranging from 150 $\Omega \cdot cm^2$ in the early portion to nearly six times higher in the later segments.[239,247]

CALCIUM

The distal convolution reclaims 4% to 8% of the calcium filtered by the glomerulus[70-72] (see Fig. 7.4). Calcium is reabsorbed by an active transcellular process, allowing the reabsorption of the ion against a lumen-negative transepithelial voltage gradient.[72,248] The Transient Receptor Potential Vanilloid 5 (TRPV5) is the main uptake mechanism for calcium in the distal convolution.[249,250] TRPV5 is part of the multigene TRP superfamily, which encodes a membrane protein that assembles into a homotetrameric ion channel. TRPV5 was cloned from primary rabbit CNT and CD isolates but only localized to the CNT in rabbit.[249,251] In mice, TRPV5 is expressed in the DCT2, CNT, and cortical CD; however, its expression in humans remains to be determined[231,252,253] (see Fig. 7.7A and 7.7B). Differences in this segmentation have been described between species, with the DCT2 segment being absent in rabbits, shortened in humans, and more prominent in mice and rats.[231,253] Massive renal calcium wasting, osteopenia, and compensatory intestinal hyperabsorption are observed in mice with targeted ablation of *Trpv5*.[254] These *Trpv5*-deficient mice showed normal proximal tubular calcium handling, but calcium reabsorption along the distal convolution was markedly altered (see Fig. 7.7C). As potassium secretion along the distal convolution increased, potassium concentration was used as an indication of distance at various puncture sites along the distal convolution. Wild-type mice display a reduction in luminal calcium delivery along the length of the micropuncturable distal convolution, in accordance with tubular reclamation of calcium along the segment. However, *Trpv5*-deficient mice showed persistently elevated calcium concentrations along the length of the distal convolution, consistent with a failure to reabsorb calcium from the DCT/CNT and indicating that this is the segment responsible for their hypercalciuria[254] (see Fig. 7.7C). In humans, pathogenic variants that cause loss of function have not been identified, but a missense SNP, albeit rare, has been described to correlate with kidney stone disease.[255] In addition, TRPV6 may provide an additional contribution to this transport, but animals lacking *Trpv6* have not been described to display significant calcium wasting[256,257] (see Fig. 7.7A and 7.7B).

Calbindin-D_{28K} is expressed along with TRPV5 in the distal convolution (see Fig. 7.7A and 7.7B). Calbindin-D_{28K} is an intracellular protein that can bind calcium with high affinity. It aids in transferring calcium across the cell and buffers intracellular calcium concentrations.[258] This is thought to contribute to high vectorial flux across the cell, without substantially elevating intracellular calcium concentrations. Furthermore, a high intracellular calcium concentration also inhibits TRPV5 activity.[259] However, hypercalciuria has not been observed in animals with deletion of the *Calb1* gene encoding Calbindin-D_{28K}.[260] Whether the absence of hypercalciuria results from compensation from other calcium-binding proteins expressed in the segment remains to be determined. Insights from a number of different knockout mouse studies suggest this may be the case. The targeted ablation of the *Vdr* gene leads to large disturbances in calcium balance, including a 90% reduction in Calbindin-D_{9K} in kidney; however, this is largely without changes in Calbindin-D_{28K}. Double knockout mice with *Vdr* and the *Calb1* gene had more severe hypercalciuria and mineral disturbances in comparison with *Vdr*-deficient animals.[261] Further, the deletion of both the *S100g* gene encoding Calbindin-D_{9K} along with *Calb1* does not result in a change in serum calcium. However, it has not been reported whether

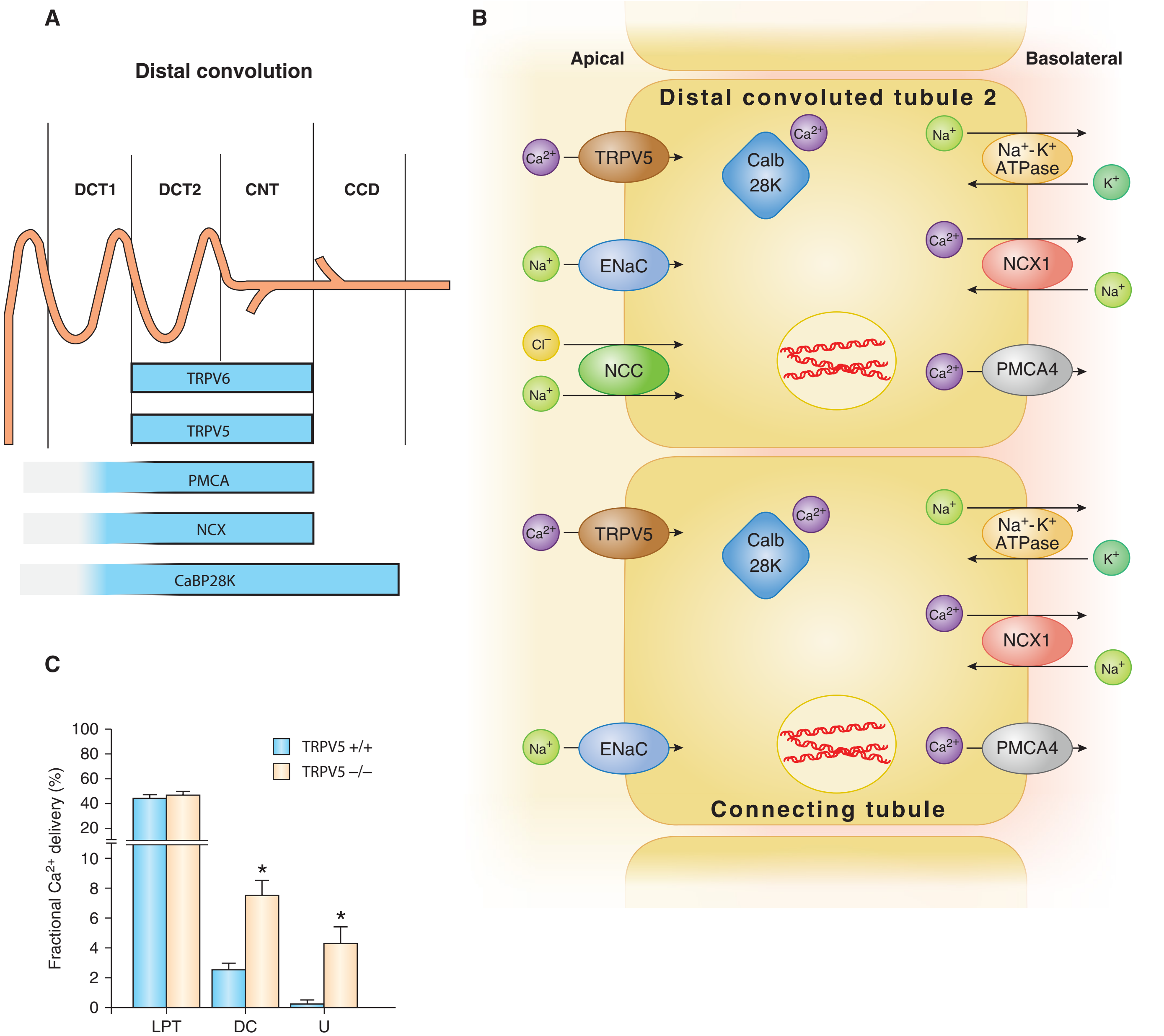

Fig. 7.7 Tubular transport of calcium in the distal convolution. (A) Distribution of calcium channels and transporters along the distal convoluted tubule (*DCT1* and *DCT2*), connecting tubule *(CNT),* and cortical collecting ducts (*CCD*). (B) Apical calcium entry is facilitated by the TRPV5 channel, which is predominantly expressed in the DCT2 and CNT. The binding of calcium to the buffering protein calbindin-D_{28K} (Calb28K) facilitates the transport of calcium to the basolateral membrane. Calcium is extruded by either the sodium/calcium exchanger 1 (NCX1) or a plasma membrane calcium-dependent ATPase (PMCA4). The sodium chloride cotransporter (NCC) is expressed in DCT1, while DCT2 expresses both NCC and the epithelial sodium channel (ENaC). Conversely, the CNT cells exclusively express ENaC. (C) Role of TRPV5 investigated by micropuncture of kidneys from TRPV5 knockout mice. The figure shows fractional Ca^{2+} delivery to micropuncture sites in the late proximal tubule *(LPT)* to sites located along the distal convolution (DC) from early to late DC (as localized using tubule K^+ concentrations) and to the urine. Deletion of TRPV5 in mice prevents Ca^{2+} reabsorption along the DC. (Adapted from Hoenderop JG, van Leeuwen JP, van der Eerden BC, et al. Renal Ca^{2+} wasting, hyperabsorption, and reduced bone thickness in mice lacking TRPV5. *J Clin Invest.* 2003;112[12]:1906–1914.)

hypercalciuria develops.[262] In addition, other yet-to-be-identified calcium-binding proteins could also compensate for this function in the distal nephron.

While the electrochemical gradient drives the influx of calcium from the lumen into the cell, calcium needs to be removed from the cell against this gradient. This is largely facilitated by the plasma membrane calcium ATPase PMCA4 and the sodium/calcium exchanger type 1 (NCX1)[238,263,264] (see Fig. 7.7A and 7.7B). They drive the secretion of calcium across the basolateral membrane by use of ATP or via a secondarily active process dependent on the sodium gradient, respectively. The PMCAs are high-affinity, low-capacity pumps that maintain low calcium concentrations inside the cytosol, while NCXs are low-affinity, high-capacity carriers. Isolation of tubules from the distal convolution from mice revealed that sodium-dependent basolateral extrusion facilitated approximately half of transcellular efflux with low vertorial flux.[265] Targeted deletion of the *Atp2b4* gene encoding PMCA4 does not lead to noticeable alterations in calcium balance, suggesting that NCX1 is the predominant efflux mechanism in this segment.[266] Furthermore, NCX2 may also contribute to basolateral calcium efflux, although the exact contribution of this exchanger has not been defined.[267] The expression of NCX1 is highest in the DCT2 and CNT region of the distal convolution, with lower expression of both PMCA4 and NCX1 in the DCT1. However, an apical calcium channel enabling calcium influx has not been identified in the DCT1[231,264] (see Fig. 7.7A).

MAGNESIUM

Detailed micropuncture experiments have demonstrated that 5% to 6% of the filtered magnesium is reabsorbed along the distal convolution.[71,80,81] The measurement of magnesium delivery to early and late puncture sites along the distal convolution of the same tubule suggests that the early convolution, which likely represents the DCT, is where the majority of magnesium is reabsorbed.[81] The luminal concentration of magnesium ranges between 0.2 and 0.7 mM in this segment,[268] whereas intracellular concentrations of magnesium are between 0.2 and 1.0 mM.[269,270] As such, the membrane potential is likely a key contributor to determining the apical entry of magnesium.[271,272] Inward transport of magnesium has generally been assumed to be mediated by the Transient Receptor Potential Melastin 6 channel (TRPM6),[273] which is a channel permeable to divalent cations. Patients with mutations of TRPM6 suffer from the autosomal recessive disorder hypomagnesemia with secondary hypocalcemia (HSH). HSH is characterized by markedly impaired absorption of magnesium from the intestinal lumen.[36,274,275] This severe disease typically presents in the first few months of life with seizures or general signs of neuromuscular excitability including tetany or muscle spasm. TRPM6 is highly expressed in the large intestine of experimental animals, where the transcellular flux of magnesium is high, explaining impaired intestinal magnesium absorption in HSH patients. TRPM6 is also expressed in the apical membrane of the DCT, as identified by immunohistochemical staining[273] (Fig. 7.8). In line with these findings is the identification of a higher-than-normal fractional excretion of magnesium in HSH patients given their blood concentration of magnesium,[274,275] a finding indicative of renal tubular magnesium wasting.[276] This renal wasting of magnesium is especially pronounced in HSH patients receiving intravenous administration of magnesium.[275] In contrast to the suggestion that TRPM6 contributes to both renal and intestinal magnesium reabsorption are the findings of Chubanov and colleagues,[277] who generated transgenic mouse lines with targeted deletion of either intestinal or renal *Trpm6*. While intestinal deletion of the channel resulted in a phenotype with lower serum magnesium levels, the renal deletion of *Trpm6* using the Ksp-Cre driver line that targets the distal tubule did not yield any changes in serum magnesium or urinary magnesium excretion.[277] As the Ksp-Cre driver line appears to target

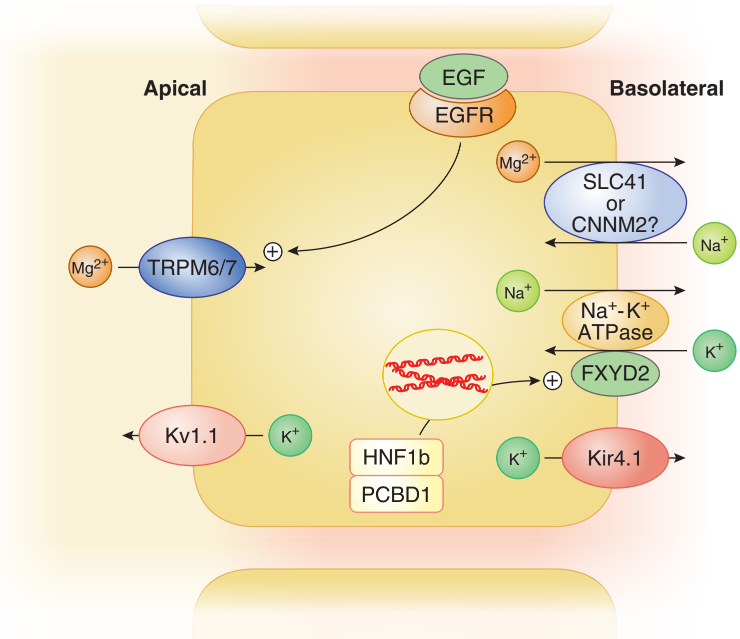

Fig. 7.8 Tubular transport of magnesium in the distal convolution. Schematic representation of the magnesium transport machinery in the distal convoluted tubule. The apical entry of magnesium is thought to occur through transient receptor potential Melastin 6 (TRPM6) and TRPM7 channels. The basolateral exit pathway for magnesium remains to be determined, but CNNM2 or SLC41A1 could contribute. The membrane voltage is adjusted by the potassium channel, Kv1.1, and influenced by the Na^+-K^+-ATPase, the accessory subunit, FXYD2, and Kir4.1, a basolateral potassium channel that recycles potassium ions that enter the cell through the Na^+-K^+-ATPase. Mutations in *HNF1B* and *PCBD1* cause hypomagnesemia and may influence FXYD2 expression. Epidermal growth factor (EGF) and epidermal growth factor receptor (EGFR) are involved in the regulation of transcellular magnesium transport in the segment.

the DCT with varying efficiency, with activity reported to be less than half in this segment, this might explain the absence of marked changes in magnesium homeostasis reported for this strain.[278] In contrast and consistent with a role for TRPM6 in renal tubular magnesium reabsorption, renal tubular *Trpm6*-deficient animals generated on a six2-cre background displayed hypomagnesemia and higher magnesium in the urine.[279] In addition to these findings, TRPM6 may not be the sole pathway for magnesium entry into the DCT, and the contribution of the close homolog TRPM7 remains to be fully delineated in this process[273,280-283] (see Fig. 7.8). However, using the Ksp-Cre driver line, *Trpm7* deletion in the kidney was not found to cause alterations in mineral balance, either with respect to urinary excretion or serum concentration.[284] Deletion of *Trpm7* using the Pax3-Cre, which excises genes in the metanephric mesenchyme, leads to an early developmental defect.[285] Additional studies with DCT-specific Cre lines are required to firmly conclude its contribution to renal magnesium transport in mice. In contrast, *Trpm7* deletion in the intestine caused a severe phenotype with disturbed mineral balance and early lethality after birth, suggesting that it does play an important role in the intestine. Two patients with pathogenic variants in the *TRPM7* gene have been reported. They display an autosomal dominant type of HSH, indicating that loss of a single allele can lead to disturbed intestinal mineral transport in humans.[286]

It currently remains to be determined in detail whether TRPM6 is required for TRPM7 to function. While some groups have shown that TRPM6 can form currents in heterologous expression systems,[287,288] others have not.[281,282,289,290] A TRPM7-like current has been detected in mouse trophoblast stem cells that express both TRPM6 and TRPM7 channels. When these cells are isolated from mice with *Trpm7*-deficiency, meaning only functional TRPM6 is left in the cell, the measured current is absent. In contrast, *Trpm6*-deficient cells that contain only TRPM7 display a reduced current amplitude but are more sensitive to magnesium inhibition, indicating that TRPM6 requires TRPM7 to function, whereas TRPM6 may alter the function of the TRPM6-TRPM7 channel complex.[277]

A number of pathogenic variants in genes expressed in the distal convolution cause hypomagnesemia. These gene defects have provided insights into mechanisms important for magnesium transport in the distal convolution.[283,291] Patients with pathogenic variants in the Na^+-K^+-ATPase γ subunit display autosomal dominant hypomagnesemia, which is secondary to urinary loss of magnesium. This disorder is often associated with hypocalciuria. Children may have muscle cramps and seizures or may be asymptomatic. This variant of hypomagnesemia was identified in two families.[292] The cause was subsequently identified as a G41R mutation in the *FXYD2* gene, encoding the γ subunit of the Na^+-K^+-ATPase. The FXYD proteins are a group of transmembrane proteins that have the ability to regulate the function of Na^+-K^+-ATPase. The localization of the γ subunit in kidney remains to be fully delineated; however, reports suggest expression in the DCT[293-296] (see Fig. 7.8). The γ subunit is important for modulating the kinetics of the Na^+-K^+-ATPase[293,297,298]; however, the protein with the pathogenic variant G41R does not localize to the plasma membrane in comparison with the wild type when expressed in cells, suggesting that Na^+-K^+-ATPase activity could be reduced.[299,300] Changes in the transepithelial voltage, resulting from this, could lower the electrical gradient for apical uptake of magnesium. However, no change in serum magnesium or urinary magnesium excretion is seen in *Fxyd2*-deficient mice or humans lacking one allele of the *FXYD2* gene,[295,299] suggesting that the G41R variant, rather than gene deletion, is causative of the phenotype. Another study has found that the γ subunit may form an inward rectifying channel, and the G41R variant changes channel properties.[301] The exact role of the Na^+-K^+-ATPase γ subunit in magnesium transport in the kidney remains to be understood in detail. The importance of adequate Na^+-K^+-ATPase activity for renal magnesium reabsorption has been highlighted in patients with de novo mutations in the Na^+-K^+-ATPase α subunit encoded by the *ATP1A1* gene, which leads to renal hypomagnesemia, refractory seizures, and intellectual disability[302] (see Fig. 7.8).

Another cause of renal magnesium wasting and hypomagnesemia is due to mutations in the transcription factor hepatocyte nuclear factor 1B (HNF1B) (see Fig. 7.8). Pathogenic variations in the *HNF1B* gene do not universally cause hypomagnesemia, but they can be part of a complex clinical picture that includes renal cysts and diabetes. In 66 patients, 44% of carriers were found to have hypomagnesemia, hypermagnesuria, and hypocalciuria.[303] The nephron segment and exact cause of the renal magnesium wasting is not clear, but studies have demonstrated that the transcription of the *FXYD2* gene is regulated by HNF1B.[303] Furthermore, pathogenic variants in the *PCBD1* gene can cause hypomagnesemia and modify HNF1B-dependent transcription of *FXYD2*[304] (see Fig. 7.8).

Patients with pathogenic variations in the *KCNJ10* gene, which encodes the inwardly rectifying potassium channel (Kir4.1, KCNJ10), have a complex syndrome that includes epilepsy, ataxia, sensorineural deafness, and tubulopathy (EAST, SeSAME). The tubulopathy manifests with hypokalemic metabolic alkalosis, hypomagnesemia, and hypocalciuria, similar to the clinical presentation of Gitelman syndrome (discussed later).[305,306] KCNJ10 is located in the basolateral membrane domains of the DCT, and the channel plays a role in recycling potassium ions across the basolateral membrane to maintain the high Na^+-K^+-ATPase activity of the DCT[238] (see Fig. 7.8). Without KCNJ10, a decrease in Na^+-K^+-ATPase activity is expected to cause cell depolarization, which in turn reduces the reabsorption of magnesium.

A rare autosomal recessive disorder that results in isolated renal hypomagnesemia has been described due to a pathogenic variant in the *KCNA1* gene encoding the Kv1.1 voltage-gated potassium channel, situated in the apical DCT membrane. These patients present with muscle weakness, cerebellar atrophy, myokymia, muscle cramps, and tetany starting from early childhood. The channel appears to play an important role in maintaining membrane voltage. Hypomagnesemia is thought to be caused by mutations in the channel, which likely decreases the inward driving force for magnesium[307] (see Fig. 7.8).

CNNM2 is highly expressed in the distal nephron and brain. Pathogenic variants in this gene result in renal magnesium loss, hypomagnesemia, seizures, and impaired intellectual development.[308,309] Although this protein appears to mediate sodium-dependent magnesium flux, its specific role in DCT magnesium transport is not fully understood[310] (see Fig. 7.8). In contrast, another member of the family, CNNM4, plays an important role in transcellular efflux of magnesium from the intestine in mice.[311] Pathogenic variants in the *SLC41A1* gene encoding a magnesium transporter result in a nephronophthisis-like phenotype in humans.[312,313] Deletion of the *Slc41a3* gene, a mitochondrial magnesium transporter, results in hypomagnesemia in mice[314] (see Fig. 7.8). However, double knockout of both the *Slc41a1* and *Slc41a3* genes does not cause exaggerated hypomagnesemia. The molecular physiology as to the etiology of these disturbances is unclear.[315] Other causes of renal-mediated magnesium wasting include pathogenic variations in the *SLC12A3* gene, which causes Gitelman syndrome and is discussed later in the section on diuretics and pathogenic variants in the *EGF* gene (see Fig. 7.8), which is discussed in the section on antineoplastic drugs.

PHOSPHATE

The contribution of the distal convolution to the reabsorption of phosphate is debated.[83] Multiple studies have been unable to detect phosphate transport in the distal convolution in animals receiving a phosphate-replete standard diet.[316] However, there could be some distal tubular phosphate reabsorption following phosphate deprivation.[81,316-318] If this occurs, the cellular mechanism remains to be defined.

TRANSPORT OF CALCIUM, MAGNESIUM, AND PHOSPHATE IN THE COLLECTING DUCT

CALCIUM

The majority of data does not support a role for the CD in the reclamation of filtered calcium; however, some reports have suggested a small amount of calcium might be reabsorbed in the collecting system. Micropuncture studies have established that the delivery of calcium to the end of the distal convolution is low, in line with the absence of significant calcium transport along the remainder of the CD.[254,319,320] Immunohistochemical staining of the kidney shows that TRPV5, NCX1, and PMCA are expressed in the CNT but undetectable in the cortical CD.[231] In contrast, mice with targeted expression of green fluorescent protein driven by the *Trpv5* promoter revealed expression along the cortical CD.[321] In the rabbit, conflicting results have been presented. While some microperfusion studies find significant calcium flux across the cortical CD,[322,323] others have failed to do so.[324,325] In the inner medullary portion of the CD in rat, microcatheterization revealed a gradual but significant decline in calcium along its length.[326] Therefore the overall contribution of the CD to calcium transport seems insignificant, but it remains to be determined whether the segment contributes a small amount of calcium transport.

MAGNESIUM

No magnesium reabsorption has been detected in the collecting duct, consistent with transcellular flux occurring primarily in the DCT.[324] In addition, microcatheterization has been unable to identify magnesium reabsorption in the CD in the inner medulla.[326]

PHOSPHATE

While some microcatheterization studies have described a small amount of phosphate transport in the collecting duct under some conditions, the contribution appears small and the potential molecular mechanism mediating this is not well understood.[327,328]

REGULATION OF RENAL CALCIUM, MAGNESIUM, AND PHOSPHATE TRANSPORT

Calcium, magnesium, and phosphate transport in the kidney is regulated by a range of hormones, changes in dietary intakes, drugs, and other factors (see Tables 7.2, 7.3, and 7.4). The sections below detail some of the main hormonal regulatory systems and commonly prescribed drugs with effects on the transport of calcium, magnesium, and phosphate in the kidney.

HORMONAL REGULATION OF RENAL CALCIUM, MAGNESIUM, AND PHOSPHATE TRANSPORT

PARATHYROID HORMONE

Lowered blood calcium levels relieve the CaSR-dependent inhibition of PTH release from the parathyroid, which results in the secretion of PTH into the circulation. The pre-prohormone PTH is processed and ultimately released into the circulation as an 84-amino acid hormone, with the first 34 amino acids sufficient to confer biological activity via binding to a PTH receptor, PTH1R or PTH2R. PTH1R is the main receptor responsible for the renal actions of PTH and is expressed along the renal tubular epithelium, as well as in bone, whereas the PTH2R expression is restricted to extrarenal sites including the pancreas, testis, central nervous system, and placenta.[329] In situ hybridization and single-cell RNA sequencing have primarily found *PTH1R* to be expressed in podocytes and along the PT, TAL, and distal

Table 7.2 Factors Altering the Reabsorption of Calcium in the Kidney

Calcium Reabsorption Increased By	Calcium Reabsorption Decreased By
1,25-dihydroxyvitamin D	Aminoglycosides
Amiloride	Androgens
Calcitonin	Calcimimetics
Estrogen	Calcineurin inhibitors
Fibroblast growth factor 23 Klotho	Hypercalcemia
Hypocalcemia	Insulin/glucose
Metabolic alkalosis	Loop diuretics
Phosphate loading	Metabolic acidosis
PTH	Phosphate depletion
PTH-related peptide	Volume expansion
Thiazides	
Volume contraction	

Table 7.3 Factors Altering the Reabsorption of Magnesium in the Kidney

Magnesium Reabsorption Increased By	Magnesium Reabsorption Decreased By
β-Adrenergic agonists	1,25-dihydroxyvitamin D
Aldosterone	Aminoglycosides
Amiloride	Calcineurin inhibitors
Arginine vasopressin	Hypercalcemia
Calcitonin	Loop diuretics
Glucagon	Magnesium loading
Insulin	Metabolic acidosis
Magnesium restriction	Phosphate depletion
Metabolic alkalosis	Prostaglandin E_2
Parathyroid hormone	Thiazide

Table 7.4 Factors Altering the Reabsorption of Phosphate in the Kidney

Phosphate Reabsorption Increased By	Phosphate Reabsorption Decreased By
Growth hormone	Atrial natriuretic peptide
Hypocalcemia	Diuretics
Insulin	Dopamine
Metabolic alkalosis	Estrogen
Parathyroidectomy	Fibroblast growth factor 7
Phosphate depletion	Fibroblast growth factor 23
Stanniocalcin 1	Glucocorticoids
Volume contraction	Hypercalcemia
	Hypokalemia
	Matrix extracellular phosphoglycoprotein (MEPE)
	Metabolic acidosis
	Parathyroid hormone
	Phosphate loading
	Secreted frizzled-related protein 4 (sFRP-4)
	Volume expansion

convolution, with little or no expression in the remainder of the collecting system.[211,330-334] Immunohistochemistry confirmed PTH1R localization to the PT, TAL, and distal convolution, where PTH1R is found in the basolateral membrane.[211,335] These segments also show enhanced adenylate cyclase activity in the presence of PTH, resulting in elevated cAMP formation.[336-340] Additionally, PTH1R has been localized to the brush border membrane of the PT in rodents.[341,342] The overall effect of PTH is to reduce urinary calcium excretion, thereby conserving calcium in the blood.

Calcium

It remains to be fully delineated how PTH affects the handling of calcium in the PT. It is well established that PTH reduces sodium reabsorption in this segment,[58,343] by inhibiting the expression and activity of NHE3.[344-350] PT sodium and calcium reabsorption are coupled. Consistent with this, micropuncture experiments in dogs revealed that PTH reduces calcium reabsorption from the segment.[58,351] However, in hamsters, the reabsorption of calcium was found to be slightly increased.[352] Overall, the effect of PTH on PT calcium handling remains to be determined.

A high abundance of PTH1R receptors is found in the TAL.[211] In human kidney, single-cell sequencing found that the *PTH1R* gene was expressed most highly in a select cluster of cells from the TAL, which also displayed a high *CLDN16* expression.[211] PTH1R expression in select cells was confirmed by immunohistochemistry in the cortical portion of the human TAL.[211] PTH strongly stimulates calcium transport from the cortical TAL,[58,165,167,203,353-355] whereas it does not appear to influence medullary TAL calcium transport.[162,167,203] The effect is mediated via the cAMP pathway as both cAMP analogs[165,355] and cAMP-elevating hormones[167,203,356] have a similar effect, namely to increase TAL calcium reabsorption. PTH increases calcium reabsorption from the cortical TAL by adjusting the paracellular permeability and transepithelial voltage.[203,353,354] When the voltage gradient is eliminated entirely, PTH has no effect on calcium transport in this segment.[353,357] In certain, but not all studies, PTH raises the lumen-positive transepithelial voltage across the TAL.[58,165,203,353-355] Furthermore, the stimulation of calcium transport in the TAL seems to be greater than what can be explained by alterations in transepithelial voltage alone, in line with PTH influencing the paracellular permeability of the shunt. This is also supported by the observation that PTH increases calcium flux across the segment when the transepithelial voltage is clamped.[353] The mechanism that underlies PTH's influence on the permeability of the paracellular shunt may be due to regulation of CLDN16 or CLDN14. The localization of CLDN16 to the tight junction was found to be altered in cell culture models by PTH via the phosphorylation of a protein kinase A phosphorylation site on serine 217 of CLDN16. This finding offers one possible pathway for PTH-dependent regulation of paracellular permeability downstream of cAMP signaling and protein kinase A activation.[358] However, it is unclear if a shuttling procedure of this kind could introduce enough CLDN16 into the junction quickly enough to enable changes in calcium permeability. Apart from its potential role in regulating CLDN16, PTH influences the expression and localization of CLND14 (as is discussed in detail later), and additional mechanisms may also contribute.[359]

When micropuncture was performed at early and late collection sites along the same distal convolution, PTH was found to nearly double calcium reabsorption.[81] Specifically, PTH promotes calcium reabsorption in the distal convolution "granular" portion that encompasses the CNT.[360] PTH administration raises cAMP levels in both primary cell cultures derived from rabbit CNT/CCD and isolated perfused rabbit CNTs.[361,362] PTH increases TRPV5 expression in the kidney, and parathyroidectomy decreases expression of both TRPV5, calbindin-D_{28K}, and NCX1, which return to baseline levels after PTH resupplementation.[363] In primary rabbit cultures from the late distal convolution, PTH both increases the expression of TRPV5, calbindin-D_{28K}, and NCX1 and increases transepithelial calcium flux.[363] When ruthenium red was added to block TRPV5 permeability, TRPV5 expression was still increased by PTH, whereas both calbindin-D_{28K} and NCX1 displayed a reduction in their expression. These findings suggest that PTH directly stimulates TRPV5 expression; however, the expression of calbindin-D_{28K} and NCX1 is regulated by calcium influx that is driven by TRPV5.[363] In contrast, in mice with distal nephron *Pth1r* deletion mediated by the *Ksp*-Cre driver line, there was a paradoxical increase in *Trpv5* expression, but the degree of *Pth1r* deletion in this segment was not determined, and these results might be the consequence of increased TRPV5 expression to compensate for calcium losses from earlier nephron segments.[359] PTH also acutely affects TRPV5 by increasing the channel open probability and caveolae-mediated endocytosis, respectively, via protein kinase A– and protein kinase C–dependent mechanisms.[364-366] Overall, PTH increases the reabsorption of calcium along the TAL and distal convolution by modulating paracellular and transcellular transport in these segments, respectively.

Magnesium

PTH stimulates magnesium reabsorption in the TAL and DCT.[81,367,368] The mechanism mediating increased magnesium reabsorption from the TAL is likely via altered transepithelial potential difference across the TAL and increased permeability, given the effect of these hormones on calcium permeability in this segment and the known permeation properties of CLDN16 and CLDN19 to magnesium. Micropuncture data support increased magnesium reabsorption from the distal nephron in response to PTH[367]; however, whether PTH alters the expression or activity of TRPM6 is unknown.

Phosphate

PTH increases the urinary excretion of phosphate[58,343] (Fig. 7.9). PTH reduces proximal tubule phosphate reabsorption by decreasing the brush border membrane abundance of SLC34A1 and SLC34A3, as well as PIT2.[147] The mechanisms whereby this occurs have been studied in detail for SLC34A1.[369] PTH activation of the PTH1R expressed in the basolateral membrane of the PT results in the activation of the protein kinase A–dependent pathway and downstream extracellular-signal-regulated kinases (ERK) signaling leading to the phosphorylation of NHERF1.[370,371] NHERF1 is a scaffolding protein important for tethering SLC34A1 to the brush border membrane (discussed earlier). PTH-dependent downstream phosphorylation of NHERF1 leads to dissociation from SLC34A1 and subsequent endocytosis via clathrin-coated pits, ultimately resulting in its degradation in lysosomes.[372-374] When transfecting variants in NHERF1 from patients with nephrolithiasis or bone mineralization defects, a potentiated response in cAMP formation is seen after PTH administration to cells expressing the *NHERF1* variants. Furthermore, while administration of PTH inhibited phosphate uptake in all transfected cells, PTH had a greater ihibitory effect on phosphate transport in cells transfected with the *NHERF1* variants than the wild-type variant.[156]

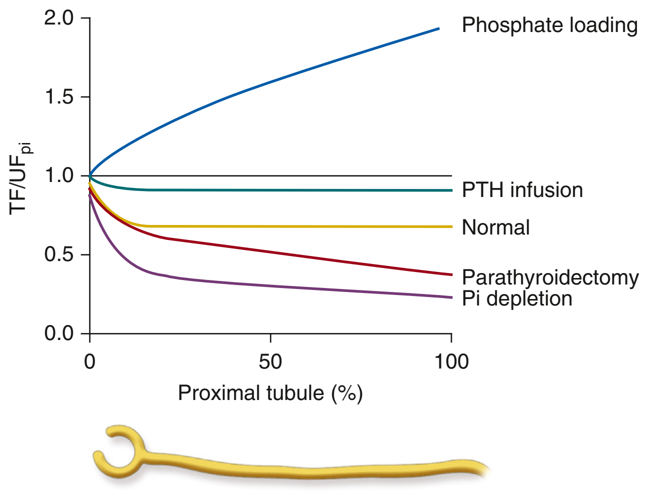

Fig. 7.9 Changes in phosphate absorption along the proximal tubule (PT) in various conditions. The effects of dietary phosphate loading or deprivation, parathyroid hormone *(PTH)* infusion, or parathyroidectomy on phosphate absorption along the PT. *Proximal tubule %,* Distance along the PT as a percentage of total length; *PTH,* parathyroid hormone; *TF/UFPi*, ratio of tubular fluid-to-ultrafiltrate phosphate concentration.

In addition to this regulation via basolateral PTH1R signaling, apical receptors have also been identified in the PT. Activation of these receptors also causes internalization of SLC34A1, although this has been described to occur through a phospholipase C and a protein kinase C–dependent pathway.[375] Activation of this pathway via PTH1R in the apical membrane also activates downstream ERK signaling, releasing SLC34A1 from NHERF1 and hence its tethering to the brush border membrane.[342,376,377] While PTH also causes internalization of SLC34A3, unlike SLC34A1, it does not undergo subsequent degradation.[147,378] The precise molecular mechanisms underlying the internalization of SLC34A3 have not been completely elucidated, although there is evidence suggesting the involvement of PKC.[379] There is limited knowledge regarding the impact of PTH on PiT-2, even though the hormone decreases the abundance of the transporter at the brush border.[147]

It is not surprising then that primary hyperparathyroidism causes hypercalcemia and symptoms associated with this electrolyte disturbance including GI upset, polyuria, polydipsia, muscle fasciculations, kidney stones, or bone pain. This is typically due to parathyroid hyperplasia, adenoma, or less commonly a malignant tumor of the parathyroid glands. Conversely, primary hypoparathyroidism presents with symptoms of hypocalcemia including muscle aches/spasms, tetany, and seizures. Hypoparathyroidism is most commonly the result of damage from thyroid surgery but can be due to genetic mutations, or autoimmune disease. Due to the effect of PTH on renal phosphate excretion, phosphate levels are typically lower in hyperparathyroidism, whereas they are often high or in the high end of the normal range in hypoparathyroidism.

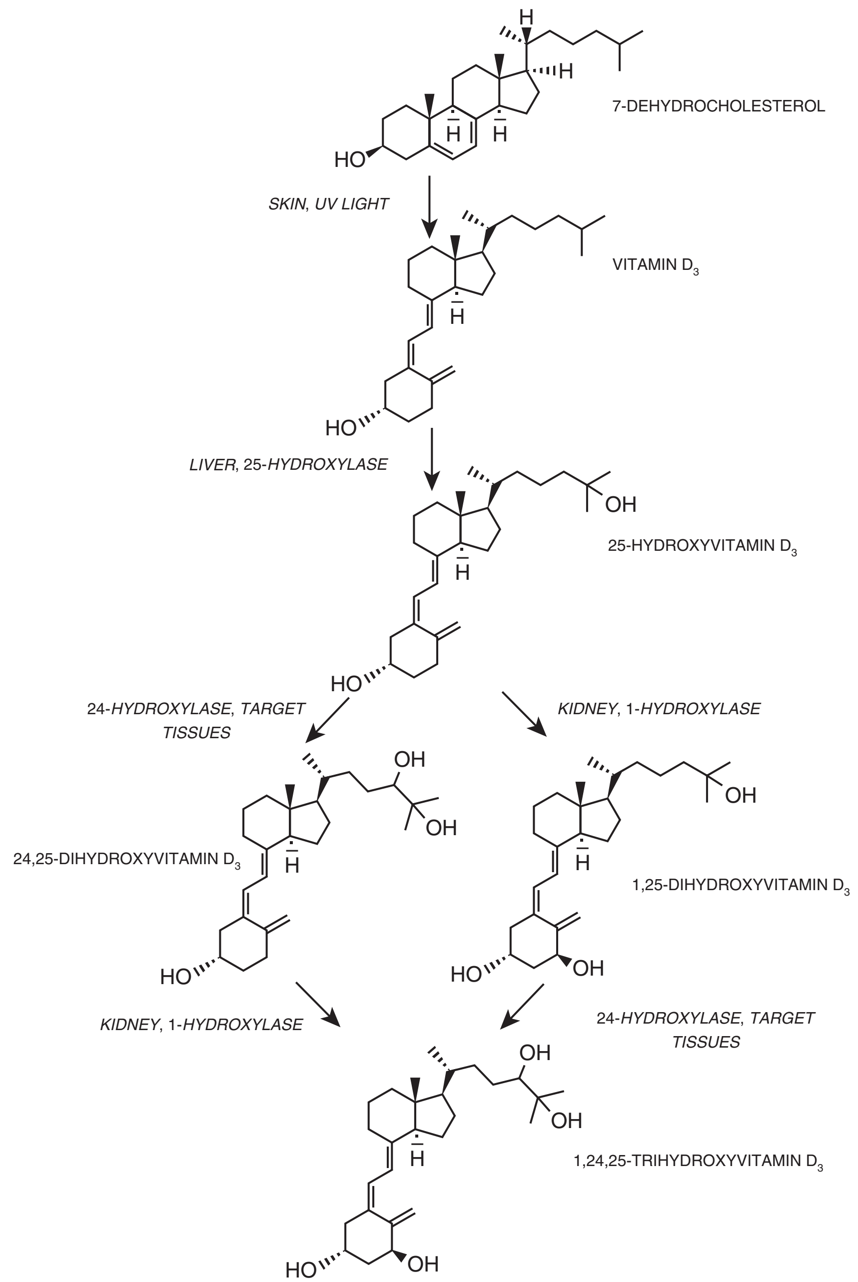

Fig. 7.10 Formation and metabolism of vitamin D3. 1-hydroxylase (1-α hydroxylase, *CYP27B1,* cytochrome P450 family 27 subfamily B member 1). 24-hydroxylase (CYP24A1, Cytochrome P450 family 24 subfamily A member 1).

1,25-DIHYDROXYVITAMIN D

1,25-dihydroxyvitamin D (calcitriol) plays a critical role in the maintenance of calcium balance.[380] Vitamin D_3 is produced in the skin. In keratinocytes, UVB light converts 7-dehydrocholesterol to pre–vitamin D_3 and this precursor is then converted to vitamin D_3 (Fig. 7.10). Vitamin D_3 (cholecalciferol) and vitamin D_2 (ergocalciferol) can also enter the circulation via dietary consumption. Vitamin D_2 is obtained from plants, whereas vitamin D_3 is found in meat and milk products. After entering the circulation, these compounds can be hydroxylated to form 25-hydroxy-vitamin D (calcifediol) by cytochrome P450 enzymes in the liver.[381,382] Subsequent conversion of 25-hydroxy-vitamin D to the active hormone, 1,25-dihydroxyvitamin D, is mediated by the cytochrome P450 enzyme CYP27B1, which is also known as the 1-α-hydroxylase (1α-hydroxylase).[380,383] 1α-hydroxylase is strongly stimulated by PTH and inhibited by 1,25-dihydroxyvitamin D, FGF23, and calcium.[384-387] The effect of 1,25-dihydroxyvitamin D is dependent on binding to the vitamin D receptor (VDR), which, in complex with the retinoid X receptor (RXR), drives gene expression via the VDR response element (VDRE) in the promoters of target genes. The VDR is present in mineral-transporting epithelia such as the kidney and intestine, as well as bone, in addition to a number of other organs, where it relays the effect of 1,25-dihydroxyvitamin D on target tissues.[380]

Calcium

1,25-dihydroxyvitamin D strongly stimulates intestinal calcium absorption[388] via the upregulation of transcellular and paracellular transport mechanisms mediated by *Trpv6* and *Cldn2*. *Cldn2* is reported to be regulated by vitamin D in the intestine.[389,390] Consistent with this, it contains a VDRE in its promoter. However, whether this is of functional significance in the kidney remains to be elucidated. Overall, 1,25-dihydroxyvitamin D seems to predominantly have a strong transcriptional effect on the calcium transport machinery in the distal convolution. Specifically, 1,25-dihydroxyvitamin D has direct effects on the expression of TRPV5 in kidney.[391] As such, the promoters of both TRPV5 and its homolog TRPV6, which is expressed more highly in the intestine, contain a VDRE.[391-393] In the TRPV6 promotor, the effect of 1,25-dihydroxyvitamin D was no longer present when the VDREs were altered by mutagenesis.[393] In *Cyp27b1*-deficient animals lacking functional 1α-hydroxylase activity, the expression of TRPV5, calbindin-D_{28K}, calbindin-D_{9K}, and NCX1 was decreased, whereas repletion of 1,25-dihydroxyvitamin D increased their expression.[394] Furthermore, in animals lacking both *Pth* and *Cyp27b1,* 1,25-dihydroxyvitamin D stimulated the expression of the renal calcium transporters TRPV5, calbindin-D_{28K}, calbindin-D_{9K}, and NCX1,[395] in line with a direct effect of 1,25-dihydroxyvitamin D on the calcium transport machinery in the distal convolution. In primary cell cultures derived from the late portion of the rabbit distal convolution, 1,25-dihydroxyvitamin D also stimulates vectorial calcium transport and the expression of select calcium-transporting genes in the segment.[361,396]

Magnesium

The effect of 1,25-dihydroxyvitamin D on intestinal magnesium absorption is less studied but increases in response to 1,25-dihydroxyvitamin D administration in some segments.[35,388,397-399] In kidneys, 1,25-dihydroxyvitamin D acutely increased magnesium excretion in thyroparathyroidectomized hamsters when infused with PTH[400] and in rats receiving a bolus of 1,25-dihydroxyvitamin D.[401] After 6 days of 1,25-dihydroxyvitamin D administration in rats, higher urinary magnesium excretion was found, in line with reduced tubular reabsorption.[402] Since serum magnesium also increased, likely due to increased intestinal absorption and potentially bone breakdown, the reduced magnesium reabsorption in kidneys could be secondary to these changes.

Phosphate

1,25-dihydroxyvitamin D also stimulates intestinal phosphate absorption. This occurs through increasing the protein expression of SLC34A2 resulting in increased transcellular phosphate absorption.[403] 1,25-dihydroxyvitamin D acutely reduces the urinary excretion of phosphate in control and parathyroidectomized rats, however, this effect was absent when rats were pretreated with cycloheximide to block de novo protein synthesis.[404] In rats with vitamin D deficiency, reductions in gene and protein expression of SLC34A1 were observed in parts of the kidney[405]; however, chronic 1,25-dihydroxyvitamin D administration was also found to reduce *SLC34A1* gene expression and protein levels.[406] Since chronic administration of vitamin D has a strong effect on intestinal phosphate absorption via SLC34A2 and mobilization of calcium from bone, the renal downregulation of SLC34A1 could also be a compensatory response to increased uptake. Often 1,25-dihydroxyvitamin D is increased with PTH, which then facilitates renal phosphate wasting.

Vitamin D intoxication or granulomatous disorders that occasionally produce the 1-α hydroxylase enzyme, and therefore elevated 1,25-dihydroxyvitamin D levels, can cause hypercalcemia and the symptoms associated with this mineral disturbance.

FIBROBLAST GROWTH FACTOR 23 (FGF23)/KLOTHO

FGF23 is a peptide hormone consisting of 251 amino acids that is produced and secreted by osteocytes and osteoblasts in response to increased levels of phosphate or 1,25-dihydroxyvitamin D in blood.[407] Proteolytic cleavage of the active hormone is essential to its biological activity. A furin protein convertase cleaves intact FGF23 at amino acid 179 into two fragments that may or may not have plasma phosphate-lowering activity.[408,409] Full-length FGF23 primarily functions to decrease the reabsorption of phosphate in the PT and inhibit the synthesis of calcitriol, but it also has effects on tubular calcium reabsorption in the distal nephron.[63,410,411] Klotho is produced at high levels in the distal convolution, and circulating klotho cleavage products can be detected in the peripheral circulation. Animals lacking the *Kl* gene encoding klotho exhibit both calcium wasting and nephrocalcinosis.[412] Klotho was first recognized for its ability to catalyze the hydrolysis of oligosaccharides linked to N-glycan, increasing the abundance of TRPV5 channels situated at the apical membrane, and thus has been implicated in calcium homeostasis.[412,413] Yet subsequent research has shown that klotho functions as a nonenzymatic scaffold that is essential for FGF23 receptor activation,[414] which is likely its more important role. In line with this, mice lacking *Fgf23* exhibit a comparable phenotype to mice lacking klotho in terms of TRPV5 membrane abundance and renal calcium wasting.[63]

Calcium

As described earlier, mice lacking the *Kl* gene or *Fgf23* waste calcium and display nephrocalcinosis, an effect secondary to the inability to retain TRPV5 in the apical membrane, thereby preventing calcium reabsorption from the distal convolution. The binding of FGF23 to its klotho coreceptor complex in this segment retains TRPV5 in the apical membrane via an ERK1/2, SGK1, and WNK4 pathway.[63] FGF23 strongly suppresses 1,25-dihydroxyvitamin D synthesis. This occurs through the activation of fibroblast growth factor receptor isoforms 3 and 4 in the PT and inducing tyrosine autophosphorylation and the stimulation of its intrinsic tyrosine kinase activity.[415] This in turn results in the activation of the MAP kinase pathway and downstream phosphorylation of extracellular signal-regulated kinase-1 and kinase-2 (ERK1/2).[387] FGF23 and klotho have a suppressive role in renal *CYP27B1* transcription through ERK1/2. FGF23 also lowers blood calcitriol levels by enhancing renal *CYP24A1* transcription, an enzyme that induces 1,25-dihydroxyvitamin D inactivation by 24-hydroxylation.[387] Consistent with this, FGF23-null mice display lower *Cyp24a1* mRNA expression in the kidney compared with wild-type mice.[416] The effects of FGF23 on magnesium homeostasis are largely unexplored.

Phosphate

Osteocytes and osteoblasts synthesize and secrete FGF23 in response to an increase in plasma phosphate or 1,25-dihydroxyvitamin D. FGF23 lowers serum phosphate in large part by reducing phosphate reabsorption from the PT and by reducing intestinal phosphate absorption through the inactivation of 1,25-dihydroxyvitamin D. In the kidney, FGF23 signals the internalization and degradation of SLC34A1 and SLC34A3 cotransporters.[418,419] This occurs via mitogen-activated protein kinase (MAPK) and serum/glucocorticoid-regulated kinase-1 (SGK-1) signaling pathways activated by FGFR 1, 3, and 4.[420-422] FGF23-FGFR signaling also downregulates transcription and translation of SLC34A1 and SLC34A3 cotransporters, leading to a decrease in their expression in the PT.[410] Moreover, PTH-induced endocytosis of SLC34A1 is abolished by inhibition of the MAPK pathway, suggesting a functional crosstalk between PTH and FGF23 signaling in the PT.[376] Hence the physiologic actions of PTH on phosphate excretion and consequent lowering of serum phosphate are enhanced by FGF23.

Increased FGF23 synthesis and secretion occurs as a result of a paraneoplastic syndrome called tumor-induced osteomalacia (TIO) or due to one of a multitude of genetic mutations,[423] in addition to occurring in kidney disease. The most common genetic cause of increased FGF23 is due to a mutation in the *PHEX* gene, for which there is now treatment with the anti-FGF23 antibody burosumab. Patients with genetic causes of increased FGF23 present with hypophosphatemic rickets.[424] Phosphatonins are a group of proteins that increase urinary phosphate excretion. The most important is FGF23. However, fibroblast growth factor 7, secreted frizzled-related protein 4 (sFR-4), and matrix extracellular phosphoglycoprotein also display this activity.[423,425]

CALCIUM-SENSING RECEPTOR

Calcium and Magnesium

The calcium-sensing receptor (CaSR) is expressed at high levels in the basolateral membrane of the TAL. In addition to this, expression at a lower abundance can also be detected along the distal convolution.[354] Some have suggested expression in the PT, but this is debated.[426] CaSR activation in TAL, as in the parathyroid, leads to a downstream cellular response that involves the formation of inositol trisphosphate, intracellular release of calcium, and inhibition of hormone-stimulated cAMP production.[427] Others have identified additional signaling mechanisms that appear to differ from the parathyroid.[53] Acute inhibition of the CaSR using calcilytics to block the activation of the CaSR leads to an increase in calcium flux in perfused TALs isolated from cortical sites. Interestingly, this appears to be a direct effect of changes in the paracellular permeability rather than a change in the transepithelial voltage.[354] In line with this observation, calcium and magnesium flux in the microperfused cortical TAL is suppressed by extracellular calcium. This phenomenon has been observed without changes in the lumen-positive transepithelial voltage or the transport of sodium chloride across the TAL.[427] The findings suggest that acute activation of the CaSR results in alterations to the permeability properties of the shunt, selectively restricting the reabsorption of divalent cations by the TAL. However, in contrast to this, studies have reported that raising calcium modifies the transcellular transport of monovalent ions in the TAL, as well as directly impacting apical ROMK activity.[428] Therefore it is probable that other mechanisms also contribute to the acute CaSR effects in the TAL, although it remains unexplained why some studies find alterations in transepithelial voltage and sodium chloride transport while others do not.

Chronic activation of the CaSR leads to additional changes in the TAL. CLDN14 is a pore-blocking claudin, unlike other claudins found in the TAL[429-431] (Fig. 7.11). In the absence of hypercalcemia or CaSR activation, CLDN14 is expressed at low levels in kidney; however, CaSR activation significantly enhances the expression of *Cldn14*.[209,429,430,432-436] *CLDN14* mutations were initially identified in individuals with nonsyndromic deafness due to the high expression of CLDN14 in the Corti organ.[437] However, no changes in calcium or magnesium balance were observed in the aforementioned patients, as well as in mice lacking the *Cldn14* gene.[431,437] Noncoding SNPs within the CLDN14 gene exhibited a significant association with kidney stone disease, hypercalciuria, and bone mineral density.[438] The expression of *Cldn14* is significantly increased by hypercalcemia or direct CaSR activation, whereas animals on high calcium diets exhibit smaller alterations in Cldn14 expression.[209,429,430,433,434] Chronic hypercalcemia caused by the administration of vitamin D derivatives results in a increase in renal *Cldn14* gene expression, ranging from 10 to 170 times higher than baseline, which in turn leads to alteration in CLDN14 protein expression.[430,433] The expression of CLDN14 markedly increases in hypercalcemic animals after a few days.[435] Importantly, the expression of *Cldn14* is directly correlated with the concentration of blood calcium, which has been utilized to determine CaSR activation in kidney.[435] The inference that this regulation is facilitated through the renal CaSR has been derived from in vitro experiments, wherein the knockdown of *CaSR* using siRNA effectively mitigates the impact of CaSR-dependent upregulation of CLDN14 expression.[429] Furthermore, it has been observed that a mere 2 hours of calcimimetic administration (compounds developed to treat secondary hyperparathyroidism that act to make the CaSR more sensitive to calcium by allosteric modification) is adequate to increase CLDN14 expression mildly.[432] In comparison, chronic activation of the CaSR in animal models through the administration of the calcimimetic, cinacalcet, enhances the expression of *Cldn14* by 40-fold.[430] Conclusive evidence supporting

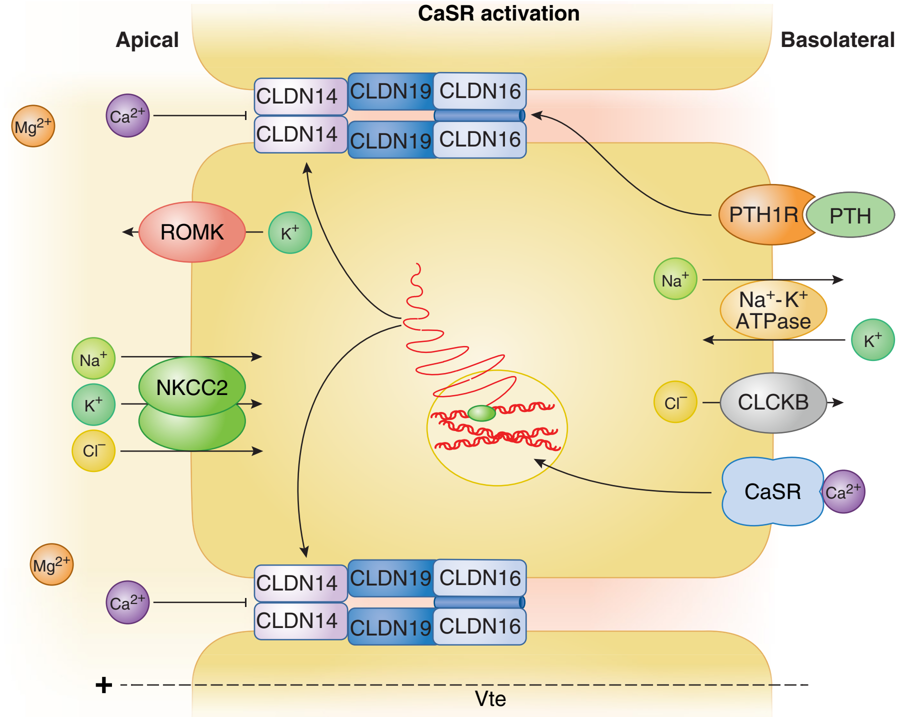

Fig. 7.11 Regulation of calcium transport in the thick ascending limb. Schematic representation of the proposed model for regulation of claudin 14 (CLDN14) by the calcium-sensing receptor (CaSR). Activation of CaSR and subsequent stimulation of CLDN14 expression occurs in the presence of hypercalcemia. CLDN14 insertion into the tight junction complex inhibits the movement of divalent cations across the segment. NKCC2 (furosemide sensitive sodium, potassium, 2 chloride cotransporter), the renal outer medulla potassium channel (ROMK), the chloride channel Kb (ClC-Kb) with its subunit Barttin, and the sodium-potassium-ATPase (Na^+-K^+-ATPase) are required for generating the lumen-positive voltage required for divalent cations reabsorption via CLDN16 and CLDN19.

a CaSR-dependent mechanism regulating CLDN14 gene expression was derived from hypercalcemic animals lacking the renal CaSR, which failed to increase CLDN14 expression. In fact, CLDN14 was largely absent from the kidney and was only preserved in a small number of TAL cells in these animals. This retained expression of CLDN14 in a few cells was likely due to the occurrence of mosaic excision in the Ksp-Cre model in the TAL used to delete the renal CaSR.[435] In addition, renal CaSR-deficient mice do not exhibit calciuresis when loaded with calcium in their drinking water, unlike their wild-type littermates,[439] which underlines the importance of the CaSR for renal regulation of urinary calcium excretion.

When overexpressed in different epithelial cell models, the pore-blocking CLDN14 decreases permeability to both monovalent and divalent cations.[429-431] The overexpression of CLDN14 leads to an elevation in transepithelial resistance and a preferential reduction in cation permeability, as indicated by a decrease in the PNa/PCl ratio, as well as the permeability to calcium. There is currently a lack of direct measurements regarding the permeability of magnesium in these models. However, considering the overall pore-blocking properties of CLDN14, it is reasonable to expect a comparable effect on the permeability of magnesium. It is worth noting that the transgenic overexpression of *Cldn14,* driven by the *Umod* promotor, in the TAL in mice augments the fractional excretion of calcium and magnesium.[432] The impact of CLDN14 on alterations in divalent cation permeability is contingent on its specific localization within the TAL. In animals subjected to a high calcium diet, CLDN14 localizes predominantly within the cortical aspects of the TAL.[209]

Staining of paraffin-embedded sections utilizing a thoroughly characterized antibody revealed that the expression of CLDN14 is limited to the OSOM and cortical TAL in animals that were subjected to hypercalcemia.[433] In contrast, animals with low or normal serum calcium levels have low or no expression of CLDN14.[433] In animals with hypercalcemia, the expression of CLDN14 is found in apical membrane domains and tight junctions but exhibits a mosaic pattern, where its expression is restricted to cells that express CLDN16.[433] Moreover, in vitro studies have reported an interaction between CLDN14 and CLDN16.[429] This observation, in combination with the mosaic localization of CLDN16 and CLDN14 to the same cells of hypercalcemic animals, offers an explanation for the predominant involvement of CLDN14 in the regulation of divalent cations, while in cell models it appears to block cations in general. These observations are corroborated by the finding that animals subjected to a high calcium diet exhibit an increased expression of CLDN14 in the cortical TAL, accompanied by a reduction in the permeabilities of calcium and magnesium.[209] In general, activation of the CaSR results in the upregulation of CLDN14, which may contribute to an increase in the excretion of divalent cations through urine. The primary regulator of this response is calcium, and while magnesium has the ability to activate the CaSR, it does not result in elevated expression of CLDN14 even in hypermagnesemic conditions in animals, suggesting that serum magnesium is not sufficiently high to activate the renal CaSR even in the hypermagnesemic range.[440]

Furthermore, apart from its potential role in regulating CLDN16, PTH may also influence CLDN14 (see Fig. 7.11). Kidney-specific *Pth1r*-deficient deletion by use of the Ksp-Cre led to a marked increase in *Cldn14* gene expression. Furthermore, the localization of CLDN14 in nonpolarized HEK cells expressing fluorescently tagged CLDN14 was observed to be shifted to cytosolic compartments upon exposure to PTH. In this model, urinary calcium wasting was blunted when *Cldn14*-deficient animals were crossed with *Pth1r*-deficient animals.[359] While these findings suggest that CLDN14 plays a crucial role in driving urinary calcium wasting, the acute effects of PTH on CLDN14 contradict the observation of minimal CLDN14 expression in TAL under normocalcemic conditions.[430,433] Hence additional research is needed to comprehend the physiologic importance of this regulatory mechanism.

In MDCK cells, activation of the basolateral CaSR reduces the reabsorption of calcium by reducing the activity of the PMCA pump via a PLC-dependent pathway.[441] Some have found CaSR in the apical site of the distal convolution, and in HEK293 cells, CaSR activation stimulates influx of calcium via TRPV5.[442] Additional studies are needed to determine the localization and effect of CaSR activation in the distal convolution.

Monoallelic activating mutations in the CaSR cause autosomal dominant hypocalcemia. This disorder is characterized by hypocalcemia and inappropriately low PTH levels.[443] Patients display hypercalciuria to a varying degree at presentation but with increasing plasma calcium develop hypercalciuria.[444] Severe activating mutations in the CaSR cause a Bartter-like syndrome with hypochloremic metabolic acidosis, as well as hypercalciuria, hypocalcemia, and suppressed PTH.[445] Conversely, inactivating mutations in the CaSR cause familial hypocalciuric hypercalcemia (FHH), a condition characterized by increased plasma calcium levels and inappropriately normal or elevated PTH[446] and the rare disorder neonatal severe hyperparathyroidism.[447]

ESTROGENS AND ANDROGENS

Calcium

Estrogen prevents osteoclastic bone resorption, and therefore osteoporosis is associated with menopause. As such, increased osteoclastic bone resorption leads to mobilization of calcium from bone and increased loss of calcium in urine. In postmenopausal women, this increased calcium loss can be counteracted by hormone replacement.[448,449] In the kidney, estrogen administration increases the abundance of TRPV5. This occurs independent of 1,25-dihydroxyvitamin D, as estrogen also increases TRPV5 expression in *Cyp27b1*-deficient and *Vdr*-deficient animals.[450,451] In *Nr3a1* knockout mice, lacking the gene for the estrogen α receptor, *Trpv5* and other calcium transporting genes in the distal convolution were significantly repressed, in line with a direct effect of estrogen on the regulation of *Trpv5*.[450] Male sex hormones likely also play a role. Orchidectomized mice exhibited reduced urinary calcium excretion, which was restored after testosterone replacement therapy. *Trpv5* and *Calb1* abundance was found to be elevated in cases of androgen deficiency, and testosterone replacement resulted in the suppression of these transporters.[452]

Magnesium

Blood magnesium was found to be inversely correlated with estrogen concentrations in menopausal women.[453] Furthermore, postmenopausal hypermagnesuria occurs and can be reduced to premenopausal levels by estrogen replacement therapy.[454] In kidney, renal *Trpm6* gene expression was found to be reduced in rats that underwent ovariectomy, and estrogen replacement resulted in the restoration of *Trpm6* gene expression, in line with an effect of estrogens in the distal convolution.[455]

Phosphate

Effects of estrogen on bone have already been mentioned. Postmenopausal women receiving estrogen supplementation have reduced blood phosphate levels compared with women without estrogen therapy.[456] In control and thyroparathyroidectomized rats, males had a higher tubular phosphate reabsorption. Furthermore, in thyroparathyroidectomized rats subjected to ovariectomy, tubular reabsorption of phosphate was increased.[457] These findings are in line with estrogen having an inhibitory effect on renal phosphate transport. In support of this, SLC34A1 gene and protein expression was found to be higher in males than females. Furthermore, in ovariectomized rats, SLC34A1 was increased compared with ovariectomized rats receiving estrogen supplementation.[458]

REGULATION OF RENAL CALCIUM, MAGNESIUM, AND PHOSPHATE TRANSPORT BY DRUGS

DIURETICS

SGLT2 inhibitors are frequently employed to treat type 2 diabetes, confer protective effects on renal function, and prevent cardiovascular events. An increased fracture rate has been associated with these compounds.[459,460] Increased serum phosphate, PTH, and FGF23 have been observed to occur with treatment,[461-463] although altered renal excretion of calcium and phosphate has not been observed.[251] It is speculated that increased PT phosphate reabsorption drives hyperphosphatemia and subsequent increases in PTH and FGF23 causing bone fragility.[463] SGLT2 inhibitors have been reported in several case series to increase serum magnesium levels when given to patients with hypomagnesemia.[464-467] This seems to be a class effect, but the mechanism is currently unknown. Fractional excretion of magnesium was reported to be reduced in two cases[464] but unchanged in another two.[466] Post-hoc analyses and meta-analyses of trials of SGLT2 inhibitors have also shown small increases in magnesium levels.[468]

Loop diuretics, such as furosemide, are primarily used in pediatric and adult patient populations for the treatment of edema. As they reduce the transepithelial voltage gradient in the TAL, they limit the reabsorption of divalent cations. This is mirrored in experimental animal models, where loop diuretics can cause urinary calcium and magnesium wasting.[187,189-191] Similar findings have also been observed in patients and healthy volunteers receiving loop diuretics.[469-473] Hypomagnesemia has been reported following loop diuretic treatment in humans; in contrast, studies on the occurrence of hypocalcemia are limited[276,474] and direct comparisons of these side effects following the administration of loop diuretics are lacking. The appearance of symptoms related to overall mineral balance may also depend on the dose and frequency of administration, as well as preexisting mineral imbalances.[251] Loop diuretics did not alter urinary phosphate excretion in healthy humans acutely.[470] Loop diuretics are discussed further in Chapter 49.

Thiazides and thiazide-like diuretics remain the most frequently prescribed hypertension medication.[475] The class of diuretics acts by blocking the NCC cotransporter.[238,476,477] In addition, they also have a net positive effect on calcium balance by reducing urinary calcium excretion,[470,478-480] an important feature of the drug that also limits hypercalciuria in kidney stone patients.[480,481] The hypocalciuric effect appears independent from changes in PTH,[480] and a similar phenomenon is observed in patients with the salt-wasting Gitelman syndrome due to pathogenic variants in the *SLC12A3* gene.[482] The hypocalciuric effect is also documented in animals following thiazide administration or genetic mouse models deficient in *Slc12a3*.[320,483-486] The hypocalciuric effect remains to be fully clarified and may result from altred tubular calcium transport in both the PT and distal convolution (see reviews for additional details[251,487]). The most explored pathway for hypocalciuria is compensatory hyperabsorption of calcium in the PT. This follows thiazide-induced volume contraction, which augments PT reabsorption of sodium and hence also calcium. As such, the hypocalciuric effects of thiazides are blunted when salt loss is compensated in thiazide-treated individuals.[480] Furthermore, volume depletion by salt restriction does not induce hypocalciuria in *Slc12a3*–deficient mice but it does in wild-type littermates.[483] Micropuncture experiments have confirmed this in animals after chronic thiazide administration[319] and in *Slc12a3*-deficient animals.[320] The hypocalciuria was still present in *Trpv5*-deficient mice following thiazide administration,[319] suggesting that the PT contributes importantly to the hypocalciuric effect. A distal mechanism to drive hypocalciuria may also exist in addition to the PT mechanism[486-489]; however, this pathway may primarily contribute in the absence of volume contraction[486] and the exact nature of this pathway remains elusive.[251,487]

In patients, chronic thiazide treatment may lead to the appearance of hypomagnesemia, albeit infrequently in some populations.[490,491] In contrast, hypomagnesemia is a characteristic feature observed in patients with Gitelman syndrome.[482] The differences in the appearance of hypomagnesemia may depend on the degree of NCC transporter inhibition.[490] The hypomagnesemia results from renal magnesium wasting. In experimental animals, both Slc12a3-deficiency and chronic thiazide administration lead to atrophy of the DCT1 segment in the distal convolution and hypertrophy of the DCT2 and CNT.[319,320,483,484,492] This may depend on the degree of inhibition and length of treatment,[492] as other studies find no significant changes in the volume of the DCT.[319,486] This segment plays a key role in renal magnesium transport, which likely explains the observed hypomagnesemia and renal magnesium losses following thiazides and Gitelman syndrome. Furthermore, direct downregulation of TRPM6 has also been reported without significant alterations in DCT morphology after thiazide administration in experimental animals.[319] Thiazide diuretics were not found to alter urinary phosphate excretion acutely.[470] Thiazide diuretics are discussed further in Chapter 49.

Potassium-sparing diuretics such as spironolactone and amiloride may have mild effects on the urinary excretion of both calcium and magnesium, acting to increase reabsorption of both in experimental animals and humans[469,490,493-506]; however, the underlying mechanisms remain to be explored in detail.[251] For further discussion on the effects of aldosterone, see Chapter 12. The potassium-sparing diuretic, triamterene, did not alter urinary phosphate excretion in healthy volunteers.[507]

AMINOGLYCOSIDES

Calcium and Magnesium

Aminoglycoside antibiotics are commonly employed for the treatment of gram-negative bacterial infections and include a number of compounds such as gentamicin, tobramycin, and neomycin. Gentamicin is frequently prescribed for children, especially in neonatal care units.[508] However, its use is limited by the occurrence of nephrotoxicity and ototoxicity.[509,510] Gentamicin can induce hypocalcemia, hypomagnesemia, and other electrolyte abnormalities due to renal wasting.[511-515] In addition, renal mineral wasting following chronic gentamicin treatment may also be attributed to renal damage. Studies from both humans and rats have demonstrated that gentamicin causes renal calcium wasting within a few hours of administration, as well as magnesium wasting in most studies, which implies acute alterations in renal mineral transport.[513,516-520] Gentamicin has generally been thought to act via the CaSR,[521] which would lead to mineral wasting after gentamicin administration. Both acutely and chronically, gentamicin had a strong effect on urinary calcium excretion in mice but did not have major effects on *Cldn14* gene expression, and calcium wasting did not differ in *Cldn14*-deficient mice.[434] Furthermore, with in vitro, it did not activate the CaSR at concentrations 25 to 50 times higher than that seen in the plasma of patients.[434,522-525] In contrast, patch clamp analysis demonstrated that gentamicin exerts a strong inhibitory effect on TRPV5 in HEK cells and that prolonged administration led to the downregulation of calcium transport proteins in the distal convolution. These findings suggest that gentamicin induces renal calcium wasting by disrupting transepithelial calcium transport rather than by CaSR activation to drive CLDN14 expression, although a contribution from the TAL through a mechanism independent of CLDN14 cannot be excluded.[434] Previous research conducted on both humans and rats has demonstrated that gentamicin can acutely alter renal magnesium excretion, although this effect is not consistently observed across all studies.[517-520,526,527] Nevertheless, no urinary magnesium losses were observed in mice acutely or chronically receiving gentamicin,[434] which may be due to species-specific effects or attributed to variations in experimental conditions between studies. As such, the observed increase in urinary magnesium excretion following chronic administration of gentamicin in some studies could be attributed to tubular injury, while other mechanisms might also be present.

Phosphate

In healthy adults and neonates, gentamicin does not increase renal phosphate excretion acutely.[517,526,527] However, in full-term infants, increased urinary phosphate excretion has been observed following chronic administration.[526] Furthermore, high-dose aminoglycoside therapy was found to cause Fanconi syndrome with hypophosphatemia in adult patients.[528] In puppies, tubular phosphate wasting increased concomitantly with the development of PT damage associated with continued administration.[529] In one study, 3 days of gentamicin administration to rats was sufficient to reduce sodium-dependent phosphate transport in renal cortical brush-border membrane vesicles.[530] Furthermore, rats exposed to gentamicin in utero and that presumably still retained gentamicin accumulated in their kidney showed lower maximal sodium-dependent phosphate transport in brush-border membrane vesicles.[531]

ANTINEOPLASTIC DRUGS (CISPLATIN AND CETUXIMAB)

Platinum-derived compounds such as cisplatin are nephrotoxic and have multiple effects on mineral balance.[532] Hypomagnesemia is most frequently encountered with treatment and is dose dependent,[533] but effects on calcium and phosphate are also seen. Mechanistically, cisplatin has been found to downregulate the expression of *Trpm6* and *Egf* in a rat model of cisplatin nephrotoxicity, potentially explaining the hypomagnesemia.[534] Furthermore, a Gitelman-like syndrome has been reported after cisplatin therapy.[535] Cisplatin can also cause hypocalcemia, often in association with hypomagnesemia. Due to the nephrotoxic effect on the PT and development of Fanconi syndrome, hypophosphatemia is also frequently observed with platinum agents.[536,537] Furthermore, mammalian target of rapamycin (mTOR) inhibitors have also been found to cause hypomagnesemia, although the exact mechanism remains to be determined.[538]

Studies with the anticancer monoclonal antibody therapy targeting the epidermal growth factor receptor showed that 34% of patients develop hypomagnesemia and 16.8% develop hypocalcemia.[539,540] The hypomagnesemia occurs as a result of renal wasting.[541,542] Similarly, isolated renal hypomagnesemia is caused by rare autosomal recessive pathogenic variants in the proepidermal growth factor gene.[542] EGF is a powerful stimulator of TRPM6, and any abnormalities in the processing of this hormone or pharmacologic inhibition of receptor signaling lead to the disruption of TRPM6 transport, resulting in the wasting of magnesium from the kidneys.[541-543]

CALCINEURIN INHIBITORS

Calcium

Calcineurin inhibitors are critical in organ transplantation but also have direct effects on mineral balance. The most commonly used calcineurin inhibitors are cyclosporine and tacrolimus, which both cause hypercalciuria.[544] As such, tacrolimus significantly impacts the expression of both TRPV5 and Calbindin-D_{28K} in kidney,[545-547] whereas cyclosporine seems to predominantly affect *Calb1* gene expression.[548] Knockout of the calcineurin regulatory subunit B α from the DCT specifically did not change TRPV5 expression; however, expression decreased when giving tacrolimus, suggesting that calcineurin activity is required in other segments, likely the CNT, to drive this response.[549]

Magnesium

Hypomagnesemia is also frequently observed in transplant patients receiving calcineurin inhibitors, resulting from excessive renal losses.[550] Hypomagnesemia and renal wasting occur in conjunction with reduced expression of *Trpm6* in rats.[546,547] A similar response has been observed for cyclosporine.[551] Deletion of the calcineurin regulatory subunit B α specifically from the DCT leads to hypomagnesemia and a marked decrease in renal TRPM6 expression.[549]

Phosphate

The effect on phosphate remains to be determined in detail. In rats chronically administered tacrolimus, SLC34A1 expression was found to be reduced in the brush border membrane.[552] Furthermore, cyclosporine inhibits phosphate flux in OK cells and reduces *Slc34a1* gene expression in weanling rats on a low-phosphate diet.[553] While wild-type mice show an increased expression of *Slc34a1* after dietary phosphate restriction, mice with deletion of the calcineurin A β gene do not show any response.[553] In contrast, in normal rats, cyclosporine acutely reduced the fractional excretion of phosphate and increased SLC34A1 protein expression in brush-border membrane vesicles.[554]

PROTON PUMP INHIBITORS

Magnesium

Proton pump inhibitors (PPIs) are commonly used to alleviate symptoms associated with chronic acid reflux and gastric ulcers. PPIs are a class of medications that act to reduce the secretion of gastric acid by inhibiting the H^+-K^+-ATPase of the P(2)-type ATPase family, located in the parietal cells in the gastric mucosa of the stomach. Importantly, hypomagnesemia has been reported as a common side effect of PPI use, often after prolonged use.[555-557] Discontinuation of treatment can restore blood magnesium concentrations to normal levels. Overall, the cause of hypomagnesemia appears to be due to altered intestinal absorption, rather than urinary magnesium wasting.[557] In fact, urinary magnesium reabsorption is increased in hypomagnesemiac patients receiving PPIs in line with a compensatory response in the kidney.[558,559]

ACIDOSIS

Calcium

Metabolic acidosis results in increased urinary calcium excretion.[560-562] This occurs via at least two different mechanisms: the release of calcium from bone[563] and changes in renal tubular calcium transport.[564] Bone is a significant reservoir for the buffers calcium-carbonate and calcium-phosphate; thus there is good rationale to implicate its role in buffering acid and data supporting this.[565] Dissolution of bone releases calcium, either as carbonate or phosphate. The net movement of calcium from bone into blood leads to excess calcium being excreted in urine, in order to reduce circulating calcium levels. Metabolic acidosis increases ionized calcium in blood, by decreasing the amount bound to albumin. Metabolic acidosis also reduces the renal reabsorptive capacity for calcium due to a direct inhibition of calcium transport within the nephron; this is evidenced by classic human studies where metabolic acidosis was induced and urinary calcium wasting observed, despite a fall in the filtered load of calcium.[560,566] The major effect of acidosis on tubular calcium reabsorption appears to be in the distal convolution, where calcium reabsorption is inhibited by metabolic acidosis.[567] This appears to be the result of altered TRPV5 expression or localization, although additional studies are needed to adequately determine this.[568-570]

Magnesium

Metabolic acidosis increases renal magnesium loss. This appears to be due to direct effects on the TAL and distal convolution.[571] The latter effect is proposed to be mediated by a reduction in *Trpm6* expression.[485]

Phosphate

Metabolic acidosis also causes urinary phosphate loss due to bone dissolution releasing the buffer calcium phosphate as described earlier.[563] In addition, acidosis may also lead to downregulation of renal phosphate transporters, although conflicting results have been found. As such, both brush border membrane phosphate transport and the gene expression and protein abundance of SLC34A1 have been found to be reduced by chronic acidosis in the rat.[572,573] In acute acidosis, brush border membrane phosphate transport and SLC34A1 protein abundance were also decreased, but *Slc34a1* gene expression remained unchanged.[572,573] Two days of acid loading increased urinary phosphate excretion in wildtype controls but not Slc34a1-deficient mice. Here SLC34A1 and SLC34A3 protein abundance increased in wild-type mice despite reduced mRNA levels of both Slc34a1 and Slc34a3.574 This supports the interpretation that acidosis primarily reuduce transporter activity rather than affecting protein expression.

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